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Research Paper

An intelligent short-term load forecasting method for virtual power plants incorporating demand response and multi-strategy optimization

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ABSTRACT

As a collaborative control platform for integrating distributed energy sources, the scheduling and power market trading strategies of a virtual power plant (VPP) heavily rely on accurate load forecasting results. However, existing short-term load forecasting methods struggle to effectively handle the nonlinear and dynamic load characteristics present in VPP scenarios. To address this, this paper proposes a short-term load forecasting method based on multi-stage optimization strategies. The method first employs variational mode decomposition (VMD) algorithm optimized by the butterfly optimization algorithm (BOA) to adaptively decompose the original load sequence, overcoming the empirical parameter dependency of traditional VMD and generating optimal intrinsic mode function (IMF) components. Subsequently, an adaptive multimodal feature fusion smoothing denoising (AMFFS) approach is applied to achieve dynamic threshold denoising of IMF components, preserving critical load characteristics while eliminating random noise. Building on this foundation, a demand response (DR) model incorporating time-delay effects (DRM-TD) is constructed, transforming dynamic factors such as electricity pricing mechanisms and user response behaviors into feature vectors. A parallel architecture combining bidirectional temporal convolutional network (BiTCN) and bidirectional gated recurrent unit (BiGRU) with a multi-head attention mechanism (MHA) is then implemented to capture temporal features and dynamically weight key characteristics. Practical case studies demonstrate that integrating electricity pricing mechanisms and user response behaviors significantly enhances forecasting accuracy. Furthermore, the successful deployment of the model on raspberry Pi edge computing devices further demonstrates its feasibility in real-world engineering applications.

1. Introduction

In the context of power systems' shift toward high penetration of renewable energy, virtual power plants (VPP) [1] have emerged as new market participants that enhance distribution grid flexibility by aggregating distributed photovoltaics [2], energy storage systems [3], and shiftable loads [4] into a coordinated source-grid-load-storage control framework. This architecture enhances renewable energy accommodation and provides flexible regulation capacity through day-ahead energy and real-time ancillary service markets [5]. In this respect, load forecasting accuracy directly determines VPP scheduling efficiency. Hence, excessive prediction errors lead to improper reserve allocation, either insufficient spinning reserves which incurs costly emergency power procurement and penalty costs, or excessive redundancy which wastes regulation resources and compresses profit margins. Both scenarios can even cause voltage violations and frequency deviations. Therefore, accurate multi-timescale load forecasting is essential for optimizing VPP operations and ensuring grid safety [6], and developing advanced forecasting methods to meet this demand has become a key research focus.

In recent years, deep learning has been widely adopted for load forecasting and other power system applications. Specifically, long short-term memory networks [7], convolutional neural networks [8], and Transformer models [9] have demonstrated remarkable success in domains such as load forecasting [10], electricity price prediction [11], and electricity theft detection [12]. However, these models struggle with small random load fluctuations caused by sudden device startups, short-term weather disturbances, and temporary demand changes [13]. These fluctuations, primarily resulting from sudden device startups, short-term weather disturbances, and temporary changes in electricity

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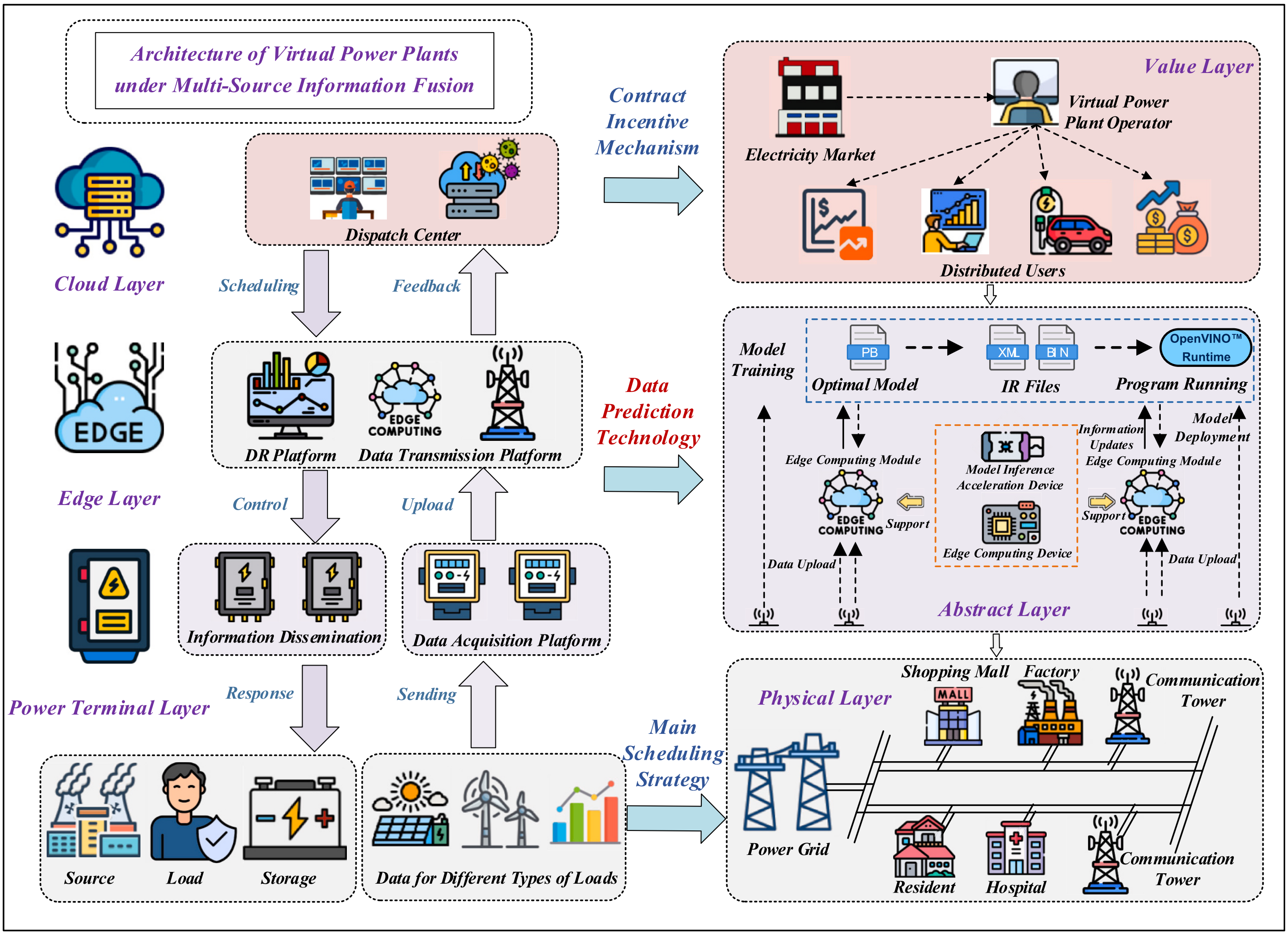


Fig. 1. Load forecasting application scenario for VPP.

demand, are characterized by high randomness and nonlinearity, with no identifiable patterns, often categorized as random noise. Random noise is essentially the result of the coupling of measurement errors, random event interference, and the uncertainty of user behavior, among other nonlinear factors. This noise leads to excessive sensitivity in prediction models to short-term disturbances, weakening their ability to capture key features such as daily and seasonal cycles, thus causing modal confusion during feature extraction [14]. This sensitivity stems from the typical non-stationarity of power load data [15]. When models attempt to fit random noise, they may lose the ability to model the essential load patterns due to overfitting. Although these fluctuations have limited impact on overall trends, their interference with the model's attention mechanism can divert focus from learning the primary features, ultimately affecting prediction accuracy.

Meanwhile, demand response (DR) technology has become a global research hotspot [16,17]. DR guides users to shift peak loads via price policies, achieving peak shaving and valley filling [18]. However, DR effectiveness depends on accurate quantitative prediction of user response behavior [19]. Without a reliable mapping between price signals and load variations, pricing policies risk supply-demand imbalances. Moreover, DR has fundamentally altered load curve formation: users are no longer passive consumers but active decision-makers influenced by price signals, environmental variables, and psychological preferences [20]. Consequently, traditional load forecasting methods that treat user behavior as static or purely weather-driven become inadequate, as they cannot capture the dynamic, price-triggered adjustments in consumption patterns. This presents a dual challenge: first, models must parse the complex mapping between price signals and user responses, identifying psychological thresholds and saturation effects; second, they must quantify the impact of response randomness on load fluctuations. Addressing these challenges requires a behavioral modeling approach that goes beyond conventional time-series forecasting.

Existing DR mechanism modeling has significant limitations. Most studies rely on economic elasticity theory or idealized assumptions, failing to capture psychological thresholds and behavioral ambiguity. For example, the hybrid model in [21] uses improved feature selection, deep restricted Boltzmann machines, and genetic optimization, but remains purely data-driven without incorporating psychological response. Similarly, the LSTM-GRU-attention model in [22] excels at time-series extraction but ignores user behavior. On randomness handling, the attention-enhanced LSTM in [23] addresses response uncertainty but lacks a systematic probabilistic framework. Studies [24,25] show that user responses under dynamic pricing exhibit significant individual and spatiotemporal differences, yet current deep learning models cannot represent these micro-level behaviors that amplify into macro-level load fluctuations. Recent works [26,27] have started breaking through traditional methods using empirical wavelet transform or particle swarm optimization, but they remain at the algorithmic level without integrating behavioral economics into deep learning.

To address the above research gaps, this paper proposes a high-precision load forecasting method that integrates adaptive denoising and time-delay DR mechanism modeling, to improve the forecasting accuracy and robustness in VPP scenarios with widespread DR implementation. The main contributions and innovations of this paper are as follows:

1) A butterfly optimization algorithm (BOA)-optimized variational mode decomposition (VMD) algorithm is proposed, which adaptively identifies the optimal decomposition layers k and penalty factor r, enhancing signal decomposition accuracy and adaptability while overcoming empirical parameter dependence in conventional VMD.
2) Building on this, an adaptive multimodal feature fusion smoothing denoising (AMFFS) method is introduced. This approach integrates a dual-entropy collaboration complexity quantification framework, a dynamic adaptive threshold decision model, and physically constrained multimodal fusion criteria to precisely identify and suppress fluctuating components in load data.
3) Based on psychological principles, a short-term load forecasting method considering time-delay effects is proposed. The demand response-incorporating time-delay effects model (DRM-TD) quantifies the dynamic impact of pricing mechanisms and user response delays, and, combined with a BiTCN-BiGRU parallel architecture and multi-head attention mechanism (MHA), achieves precise modeling of multi-source features. The feasibility of this model is validated on a raspberry Pi edge device.

The remaining structure of this paper is organized as follows: Section II presents the problem description and the proposed solution framework. Section III introduces the proposed methods. Section IV validates and analyzes the proposed methods through case studies and Section V concludes with a summary of the research content and findings.

2. Problem overview and solution architecture

In the context of the new power system architecture, VPP function as aggregators of distributed energy resources, integrating distributed photovoltaics, wind power, decentralized energy storage systems, and controllable loads. By leveraging digitalized dispatching platforms, they transform geographically dispersed resources into unified virtual generation units with collaborative control capabilities. This enables VPP to demonstrate unique advantages in enhancing renewable energy integration, participating in grid peak load regulation and frequency modulation, and engaging in electricity market transactions. The primary dispatching objective is to minimize operational costs and maximize renewable energy utilization while ensuring the balance of power supply and demand. The specific objective function can be formulated as follows:

$$\begin{cases} \min \sum_{t=1}^{T} \left(C_g(t) + C_s(t) + C_c(t) \right) \\ \quad s.t. \sum_{i=1}^{N} G_i(t) = P(t) \end{cases} \tag{1}$$

where $C_g(t)$ is the generation cost of traditional units, $C_s(t)$ is the charging and discharging loss cost of the energy storage system, $C_c(t)$ is the renewable energy curtailment penalty cost, $G_i(t)$ is the output power of the i-th generator unit, and $P(t)$ is the total system load demand.

As shown in the application scenario in Fig. 1, the achievement of this objective is highly dependent on accurate load forecasting results. These forecasts support the scheduling of the energy storage system, the formulation of controllable load adjustment strategies, and the coordinated optimization of distributed power generation.

However, traditional load forecasting methods mainly rely on historical load data and external variables such as meteorological parameters. The core assumption is that user electricity consumption behavior is entirely passively driven by the external environment, without sufficiently considering the impact of electricity market pricing mechanisms on users' active decision-making. Under mechanisms such as time-of-use pricing (TOU) and real-time pricing (RTP), users optimize their consumption behavior based on electricity costs. This leads to load curves that exhibit a strong correlation with price signals. Traditional methods, however, fail to capture the decision-driven load fluctuations due to the absence of modeling for user response behavior, which affects the economic efficiency and reliability of VPP scheduling strategies. For example, residential users might charge their energy storage devices during low-price periods and discharge during high-price periods, resulting in load curves that exhibit structural changes strongly correlated with price signals. The user energy storage optimization model can be expressed as follows:

$$\begin{cases} \max \left(\sum_{t=1}^{T} \left(P_{dis,t} \cdot P_{high,t} - P_{cha,t} \cdot P_{low,t} \right) \right) \\ s.t. \begin{cases} E_{s,t} = E_{s,t-1} + \eta_{cha} \cdot P_{cha,t} \cdot \Delta t - \dfrac{P_{dis,t}}{\eta_{dis}} \cdot \Delta t \\ E_{\min} \le E_{s,t} \le E_{\max} \end{cases} \end{cases} \tag{2}$$

where $E_{s,t}$ denotes the energy stored in the energy storage system at time t, parameters η_{cha} and η_{dis} correspond to the charging and discharging efficiencies, respectively. Additionally, $P_{cha,t}$ and $P_{dis,t}$ denote the charging and discharging power at time t.

Furthermore, in VPP operations, the electricity load data is not only influenced by users' active adjustments but is also subject to interference from multi-source random noise. This noise manifests in practical scenarios as nonlinear disturbances such as sudden device startups, short-term weather fluctuations, and random user behavior. Such disturbances demand higher robustness from the forecasting models. When models attempt to fit short-term disturbances, they may overly focus on noise, neglecting the intrinsic load patterns, and even lead to overfitting in deep learning models, resulting in significantly increased prediction errors. For example, if noise suppression is inadequately addressed, a model might mistakenly identify a sudden surge in equipment load as a long-term trend, thereby introducing bias in thermal power reserve capacity allocation and further compromising the economic efficiency and reliability of VPP dispatch strategies.

Based on the aforementioned issues, and considering the interference of random noise, the load forecasting model under VPP can be structured as a hybrid framework of multi-stage decomposition-denoising-prediction. The core idea is to extract multi-scale features through signal decomposition, dynamically reduce noise interference, and combine user response behavior modeling to significantly enhance forecasting accuracy and provide more reliable decision support for scheduling optimization. The model can be expressed as follows:

$$\begin{cases} L(t) = f(L_{rec}(t), W(t), \pi(t), \theta) + \varepsilon(t) \\ \quad L_{rec}(t) = W \cdot L_{his}(t) + b \end{cases} \tag{3}$$

where $L_{his}(t)$ denotes the historical load data at time t; $L_{rec}(t)$ represents the denoised load data after noise suppression; $W(t)$ corresponds to the meteorological data at time t; $\varepsilon(t)$ quantifies the forecasting error between predicted and actual load values. W and b represent the weight matrix and bias term for load signal reconstruction, respectively; $\pi(t)$ indicates the electricity price signal at time t, corresponding to the price difference $\Delta\omega_{p-v,t}$ defined in Section 3.3. The user response parameter set θ is defined as $\theta = \{\psi_{max,p-v,t}, \psi_{min,p-v,t}, m_{p\text{-}v,t}, n_{p\text{-}v,t}, \kappa, \rho\}$, where $\psi_{max,p-v,t}$ and $\psi_{min,p-v,t}$ denote the maximum and minimum DR thresholds under optimistic and pessimistic scenarios, respectively; and $m_{p\text{-}v,t}$ and $n_{p\text{-}v,t}$ are the lower and upper thresholds of the progressive response region, as defined in Eq. (40); κ is the optimistic response index, as defined in Eq. (42); and ρ is the response delay factor, as defined in Eq. (43). The interaction between the price signal $\Delta\omega_{p-v,t}$ and θ is governed by the logistic-type DR model formulated in Eqs. (40)–(43), which yields the actual response intensity $\widetilde{\psi}_{p-v,t}(t)$, as defined in Eq. (41).

In summary, the framework of the proposed short-term load

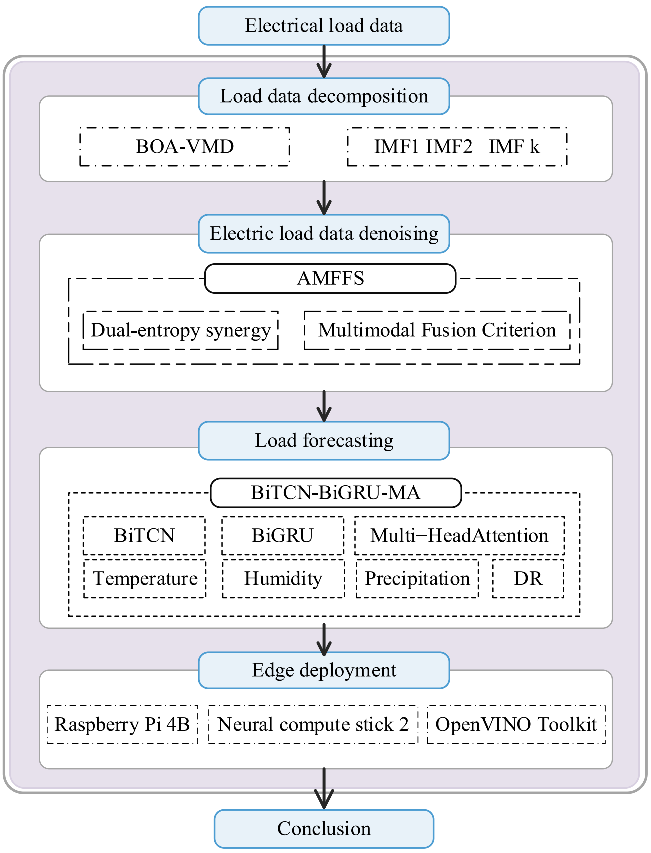


Fig. 2. Schematic of the proposed load forecasting framework.

forecasting method is shown in Fig. 2. This method adopts a multi-stage decomposition-denoising-prediction hybrid framework designed to address two major challenges faced by load data in the VPP context: random noise interference and the dynamic changes in user response behavior.

Firstly, this paper improves the VMD with a BOA to perform adaptive decomposition of the raw load series, dynamically optimizing VMD parameters to generate k optimal intrinsic mode function (IMF). This effectively separates high-frequency noise from low-frequency trends and overcomes the limitations of traditional VMD, which depends on empirical parameter settings. Subsequently, an AMFFS is applied to achieve multi-scale feature fusion and dynamic noise suppression. This method preserves key load features while eliminating random noise disturbances such as sudden device startups and short-term weather fluctuations. Based on this, a fuzzy DR quantification model considering time-delay effects is constructed, which converts dynamic factors such as electricity price elasticity coefficients and user response delay characteristics into feature vectors. This quantifies the response intensity of users to price signals, enhancing the model's ability to capture price-driven behaviors. Finally, a parallel architecture combining BiTCN and BiGRU is utilized to capture multi-scale temporal characteristics and long-term dependencies in the load sequence. An MHA is introduced to dynamically weight and fuse the feature outputs of BiTCN and BiGRU, highlighting key load features and significantly improving the accuracy and robustness of short-term load forecasting in the VPP context.

3. Research methodology

3.1. BOA-optimized VMD algorithm

Electricity load data is significantly affected by dynamic factors such as weather and holidays, presenting substantial non-stationarity. Additionally, the interplay of multi-time-scale features such as daily and seasonal cycles further complicates the load data. Traditional signal processing methods struggle to effectively address these multi-dimensional coupling characteristics. To solve this complex issue, this paper adopts the BOA-based VMD (BOA-VMD) method for the adaptive decomposition of load sequences. VMD, as a non-recursive adaptive mode decomposition technique, formulates the signal decomposition process by constructing a constrained variational problem. This approach decomposes the signal into multiple modal components with distinct central frequencies and bandwidths, enabling multi-scale feature extraction from complex non-stationary signals. The core optimization objective is to minimize the bandwidth of each modal component while ensuring that their sum can reconstruct the original signal. To solve this constrained optimization problem, VMD converts it into an unconstrained form using Lagrange multipliers and iteratively updates the modal components and center frequency set using the alternating direction method of multipliers (ADMM), ultimately obtaining the optimal decomposition result. The specific process is shown in Eq. (4):

$$\begin{cases} L(\{u_k\},\{\omega_k\},\lambda(t)) = \alpha \sum_{k=1}^{K} \left\| \left[\partial_t \left(\delta(t) + \frac{j}{\pi t} \right) {}^{*} u_k(t) \right] e^{-j\omega_k t} \right\|_2^2 + \\ \left\| \sum_{k=1}^{K} u_k(t) - f(t) \right\|_2^2 + \left\langle \lambda(t), \sum_{k=1}^{K} u_k(t) - f(t) \right\rangle \\ s.t. \sum_{k=1}^{K} u_k(t) = f(t) \end{cases} \tag{4}$$

where $u_k(t)$ represents the k-th mode component, ω_k is its corresponding center frequency, and $\delta(t)$ is the Dirac delta function. λ is the Lagrange multiplier, and α is the penalty factor.

However, the decomposition accuracy of VMD is highly dependent on the choice of decomposition layers k and penalty factor r. To overcome the limitations of traditional empirical parameter selection, this paper introduces the BOA algorithm to optimize these two critical parameters. BOA is an intelligent optimization algorithm inspired by the foraging behavior of butterfly populations. It simulates the collective behavior of butterflies and the fragrance propagation mechanism to balance global exploration and local exploitation. In the BOA algorithm, the position update of each butterfly is driven by the perceived fragrance concentration f_i, which is calculated through a nonlinear mapping of the fitness value:

$$f_i = c \cdot I_i^a \tag{5}$$

where f_i denotes the fragrance concentration perceived by the i-th butterfly, I_i represents the stimulus intensity of the i-th butterfly, c is the perception factor, and a is the power exponent parameter with a value range of [0,1].

The butterfly movement strategy consists of global and local search modes. When the fragrance concentration perceived by a butterfly exceeds the fragrance it emits, the butterfly performs a global search, moving toward the region with a higher fragrance concentration. In global search mode, the update formula for the butterfly moving toward the current global best solution g^* is shown in Eq. (6):

$$x_i^{t+1} = x_i^t + r^2 \cdot (g^* - x_k^t) \cdot f_i \tag{6}$$

where *xt i* represents the position of the i-th butterfly at iteration t, g^* is the current global best position, and r is a random number within the range [0,1].

In local search mode, the butterfly moves randomly around two randomly selected butterflies, x_j and x_k. The update formula for this local search is shown in Eq. (7):

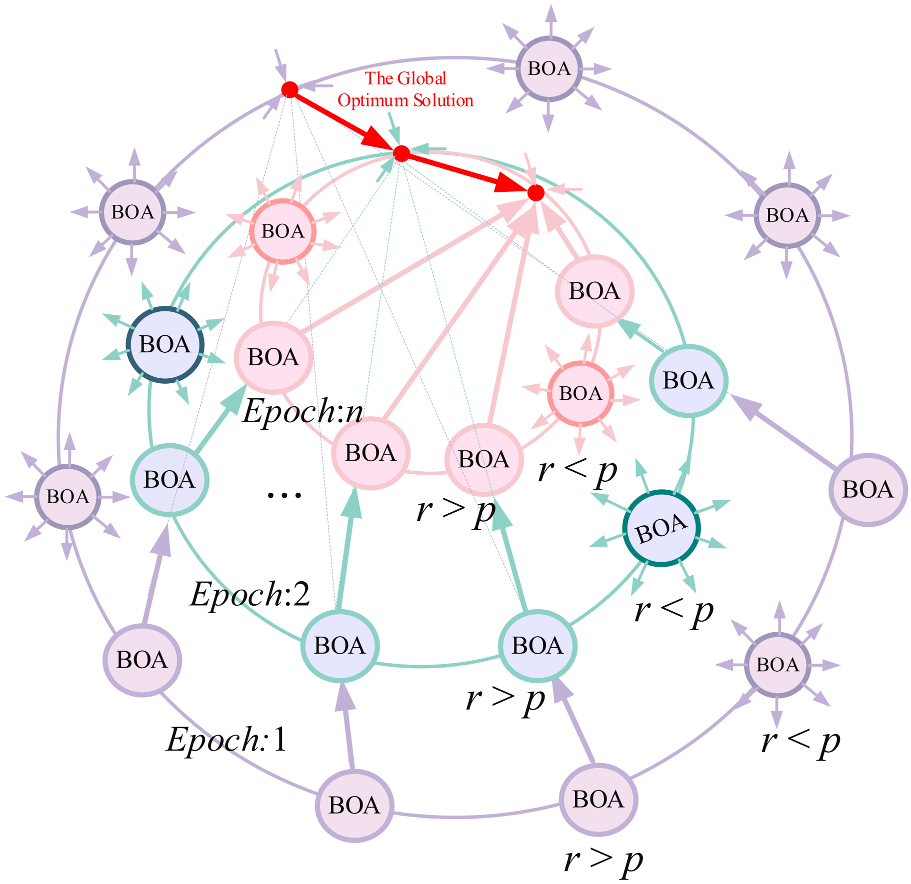


Fig. 3. Schematic of the butterfly optimization algorithm.

$$x_i^{t+1} = x_i^t + r^2\cdot\left(x_j^t - x_k^t\right)\cdot f_i \tag{7}$$

where *xt j* and *xt k* represent the positions of the randomly chosen *j*-th and *k*-th butterflies at iteration *t*, and *r* is a random number uniformly distributed.

In the BOA optimization process, global and local search strategies are dynamically switched based on a switching probability *p*. When a randomly generated number $r<p$, the butterfly performs a global search; otherwise, it switches to a local search. This dynamic switching mechanism allows butterflies to quickly locate potential regions globally and conduct fine searches locally, thus effectively balancing exploration and exploitation. The schematic of this process is illustrated in Fig. 3, which depicts how the algorithm transitions between global and local search phases. Through the BOA-optimized VMD algorithm, adaptive optimization of the decomposition layers *k* and penalty factor *r* can be achieved, enhancing the decomposition accuracy and robustness of complex non-stationary signals, providing a higher-quality data foundation for subsequent forecasting and analysis.

3.2. Adaptive multi-modal feature fusion smoothing

In load forecasting tasks, random noise has a relatively small impact on the overall load trend. Excessive focus on high-frequency fluctuations may detract from the model's ability to identify primary trends and periodic features, thus reducing forecast accuracy. To address this issue, this paper proposes an AMFFS method, which includes three core stages: fluctuation characteristic quantification, dynamic threshold decision-making, and fusion with physical constraints. The method aims to achieve collaborative optimization in signal denoising and the retention of key features through multi-scale feature decoupling.

(1) Quantification of fluctuation characteristics based on dual-entropy synergy.

Entropy, as a core metric for measuring signal uncertainty, needs to balance dynamic responsiveness and robustness. To select the appropriate entropy for quantifying signal fluctuation characteristics, Table 1 compares the applicability of various entropy measures in power load data.

From Table 1, it is clear that PE performs excellently in load dynamic response and fluctuation sensitivity but is sensitive to data loss. On the

Table 1
Comparison of applicability of different entropy measures in power load data.

Criterion	Permutation entropy (PE)	Sample entropy (SampEn)	Shannon entropy (SEn)	Fuzzy entropy (FuzzyEn)	Multi-scale Entropy (MSE)
Load dynamic response	√	Δ	×	Δ	×
Sensitivity to fluctuations	√	√	×	√	√
Non-stationary adaptation	√	√	×	√	√
Computational efficiency	√	Δ	√	×	×
Physical interpretability	√	√	Δ	Δ	Δ
Insensitivity to data loss	Δ	√	×	×	×

where √: Fully satisfied; Δ: Partially satisfied; ×: Not satisfied.

other hand, SampEn exhibits strong robustness to data loss, compensating for the weaknesses of PE. Therefore, this paper constructs a dual-modal evaluation model by normalizing and weighting the fusion of PE and SampEn, resulting in a comprehensive complexity score. Specifically, PE captures short-term dynamic characteristics by analyzing the permutation pattern distribution of the time series, while SampEn strengthens the long-term trend analysis using the Heaviside function. The normalized composite score provides a quantitative basis for subsequent threshold decision-making. The steps are as follows:

1) Permutation entropy calculation.

Permutation entropy analyzes the permutation pattern distribution of a time series without assuming that the data follows a specific distribution, thus capturing dynamic characteristics. This feature makes it highly robust to noise, especially for complex signals such as power load data. The specific calculation formula is shown in Eq. (8):

$$\begin{cases} X_m^k(t) = [E_k(t), E_k(t+\rho), \ldots, E_k(t+(m-1)\tau)] \\ PE_k = -\dfrac{1}{\ln m!}\displaystyle\sum_{i=1}^{m!} p(\pi_i)\ln p(\pi_i) \\ \pi_i = rank(X_m^k(t)) \end{cases} \tag{8}$$

where $E_k(t)$ is the original time series, m is the embedding dimension length, ρ is the delay parameter, and $Xk\ m(t)$ is the probability of permutation patterns.

2) Sample entropy calculation.

Sample entropy quantifies the recurrence of similar patterns within a time series, with higher values indicating higher complexity. The calculation formula is given in Eq. (9):

$$\begin{cases} \Phi^m(r) = \dfrac{1}{N-m}\displaystyle\sum_{i=1}^{N-m}\left[\dfrac{1}{N-m-1}\sum_{j\neq i} H(r - \|X_m(i) - X_m(j)\|_\infty)\right] \\ SE_k = -\ln\left(\dfrac{\Phi^{m+1}(r)}{\Phi^m(r)}\right) \end{cases} \tag{9}$$

where N denotes the number of data points in the original time series, $\Phi^m(r)$ represents the probability that m-dimensional embedding vectors are similar within threshold r, and $H(\cdot)$ is the Heaviside function.

3) Composite score construction.

By normalizing Eqs. (8) and (9), the normalized composite complexity score is obtained, as shown in (10):

$$S_c = a\frac{PE_k}{max(PE)} + (1-a)\frac{SE_k}{max(PE)} \tag{10}$$

where a is the weight coefficient set to 0.5 in this paper.

The S_c effectively quantifies the non-stationary features of the signal and provides data support for threshold decision-making. However, entropy-based complexity assessment is inherently a first-order statistical approach, which may not fully decouple the mixed multi-source features in the signal. For instance, high-complexity components may include transient responses triggered by sudden load events, while low-complexity components may still contain low-frequency disturbances from measurement errors. This feature mixing originates from the non-stationary nature of load signals, requiring a decision mechanism enhanced by physical constraints for precise separation. To address these challenges, this section proposes a dynamic adaptive threshold decision method to overcome the limitations of traditional approaches in feature decoupling and integration of physical constraints.

(1) Dynamic adaptive threshold decision.

Based on the multi-scale complexity features provided by the composite entropy index, this section employs a probability distribution-based adaptive threshold decision method to screen IMF components, ensuring that only valid components are passed to the next processing stage. The steps are as follows:

First, the initial threshold decision is constructed using the composite complexity score S_c calculated by Eq. (10), with boundary thresholds determined by the percentile method. This stage removes significantly invalid components, as shown in Eq. (11):

$$\begin{cases} \Omega_{pre} = \{IMF_k | C_k \in [\lambda_l, \lambda_u]\} \\ \lambda_l = Q_{0.15}(C), \quad \lambda_u = Q_{0.95}(C) \end{cases} \tag{11}$$

where $C_k < \lambda_l$, the component is an over-smoothed component; where $C_k > \lambda_l$, the component is a strong noise component.

Next, probability distribution modeling is conducted on the pre-screened components, and the Akaike Information Criterion (AIC) is employed to select the optimal distribution type, as shown in Eq. (12):

$$\mathrm{AIC}_i = 2k_i - 2\ln(L_i) \tag{12}$$

where k_i denotes the number of parameters for the i-th distribution, L_i represents the log-likelihood function value.

Finally, an adaptive threshold function is calculated based on the fitted distribution, as shown in Eq. (13):

$$T_h = \begin{cases} e^{(\mu+\lambda\sigma)} \\ \mu + \lambda\sigma \end{cases} \tag{13}$$

where T_h is defined as $e^{(\mu+\lambda\sigma)}$ when the distribution is a log-normal distribution, and it is defined as $\mu + \lambda\sigma$ for any other type of distribution, μ represents the mean of the probability distribution, σ denotes the standard deviation of the probability distribution, and λ is the confidence coefficient.

(2) Multi-modal fusion criterion with physical constraints.

Traditional pure data-driven methods lack explicit modeling of the physical characteristics of power load data, making it difficult to balance system constraints and evolutionary laws during the smoothing process. After the adaptive threshold screening, this section proposes a multi-dimensional fusion criterion incorporating periodic features, long-term trends, energy distribution, and high-frequency suppression. The specific formulation is outlined as follows:

1) Main frequency periodicity criterion.

To preserve daily periodic features in power load data, the main frequency periodicity criterion explicitly constrains daily variations in the load signal, ensuring the integrity and accuracy of the data. For the k-th IMF obtained by VMD, its amplitude spectrum density function $S_k(f)$ is defined as:

$$S_k(f) = |X_k(f)|^2 \tag{14}$$

where $X_k(f)$ denotes the Fourier transform of the k-th IMF component, and $|X_k(f)|$ represents its amplitude. The daily periodic frequency domain is defined by Eq. (15):

$$\Omega_{\mathrm{day}} = \left\{ f \in R^+ \middle| \frac{1}{f} \in [T_{\mathrm{day}} - \Delta T, T_{\mathrm{day}} + \Delta T] \right\} \tag{15}$$

Therefore, the daily periodic feature criterion is formulated as Eq. (16):

$$\mathbb{I}_{cycle}(k) = \begin{cases} 1, \exists f \in \Omega_{day}, s.t. S_k(f) > \upsilon \cdot max(S_k) \\ 0, \ \text{otherwise} \end{cases} \tag{16}$$

where υ is the significance threshold, typically set between 0.1 and 0.3, which controls the minimum relative amplitude strength.

2) Long-term evolution retention criterion.

To maintain long-term evolutionary characteristics in the power load signal, a trend retention criterion based on VMD modal structure is introduced. The first two low-frequency IMF components are retained as the trend carriers, as shown in Eq. (17):

$$\mathbb{I}_{\text{trend}}(k) = H\left(k_{cutoff} - k\right) \tag{17}$$

where $H(\cdot)$ denotes the Heaviside step function, and k_{cutoff} is the low-frequency cutoff order, set to 2.

3) Energy dominance criterion.

To preserve the integrity of dominant components in the signal and avoid distortion from excessive component removal, the energy dominance criterion is introduced:

$$\mathbb{I}_{\text{energy}}(k) = \frac{\|\text{IMF}_k\|_2}{\sqrt{\sum_{i=1}^{K} \|\text{IMF}_i\|_2^2}} \tag{18}$$

where IMF_k denotes the k-th IMF, and $||\cdot||_2$ represents the Euclidean norm.

4) High-frequency noise suppression criterion.

To accurately identify and suppress high-frequency fluctuations in the IMF components, a dual constraint criterion based on time-frequency energy distribution and spectral curvature is used. Specifically, for the k-th order IMF component, the Wigner-Ville distribution (WVD) is employed to characterize its time-frequency energy distribution. The WVD provides a high-resolution representation of signal energy in both time and frequency domains, forming the basis for subsequent high-frequency oscillation detection. The mathematical formulation is defined as:

$$W_k(t,f) = \int_{-\infty}^{+\infty} x_k\left(t+\frac{\tau}{2}\right)x_k^*\left(t-\frac{\tau}{2}\right)e^{-j2\pi f\rho}d\tau \tag{19}$$

where $x_k(t)$ denotes the k-th IMF component and ρ represents the time delay variable.

A composite detection function is constructed based on the energy distribution, formulated as:

$$\mathbb{I}_{\text{noise}}(k) = \iint_{\Omega_{\text{noise}}} W_k(t,f)dtdf \tag{20}$$

where Ω_{noise} is the fluctuation area, and its specific form is shown in Eq. (21):

$$\Omega_{\text{noise}} = \left\{ (t,f)|f > f_c, \frac{\partial^2 S_k(f)}{\partial f^2} > \gamma \right\} \tag{21}$$

where f_c denotes the critical frequency threshold, above which components are considered candidate fluctuations. γ represents the spectral curvature constraint, used to filter high-frequency fluctuations with abrupt spectral variations.

The output results of individual criteria are fused via a tensor product operation to construct a multi-criteria fusion decision mechanism, as shown in Eq. (22):

$$\mathscr{D}(k) = \mathbb{I}_{\text{cycle}} \otimes \mathbb{I}_{\text{trend}} \otimes \mathbb{I}_{\text{energy}} \otimes (1 - \mathbb{I}_{\text{noise}}) \tag{22}$$

Based on this, the filtered retained component set is determined by the decision mechanism, as shown in Eq. (23):

$$\Gamma_{retain} = \{IMF_k | \mathscr{D}(k)\ \} \tag{23}$$

The reconstructed signal $\widehat{x}(t)$ is obtained by summing the filtered IMF components, as defined in Eq. (24):

$$\widehat{x}(t) = \sum_{k=1}^{\Gamma_{retain}} \text{IMF}_k(t) \tag{24}$$

(3) Multi-dimensional denoising evaluation.

To comprehensively evaluate the denoising performance of the AMFFS method, this paper proposes three assessment metrics: energy preservation rate, residual 3σ criterion, and spectral consistency, which jointly construct a multi-scale validation framework from three dimensions: energy fidelity, statistical robustness, and frequency domain integrity.

1) Energy retention rate.

The energy retention rate η measures the proportion of energy retained in the denoised signal relative to the original signal:

$$\begin{cases} \|x\|_2^2 = \sum_{t=1}^{N} x(t)^2 \\ \|\widehat{x}\|_2^2 = \sum_{t=1}^{N} \widehat{x}(t)^2 \\ \eta = \dfrac{\|\widehat{x}\|_2^2}{\|x\|_2^2} \times 100\% \end{cases} \tag{25}$$

where $x(t)$ denotes the original signal, $\widehat{x}(t)$ represents the smoothed signal.

2) Residual 3σ criterion.

The residual 3σ criterion ensures that the residual is within 3 standard deviations, guaranteeing statistical robustness during denoising:

$$\begin{cases} \epsilon(t) = x(t) - \widehat{x}(t) \\ |\epsilon(t) - M_\epsilon| < k \cdot IQR_\epsilon \forall t, \end{cases} \tag{26}$$

where $\epsilon(t)$ is the residual at time step t, M_ϵ is the median of the residual sequence, IQR_ϵ is the inter-quartile range of the residuals, the coefficient k is determined based on empirical quantiles of the residual data, with the calculation formula given in Eq. (27):

$$k = \frac{max\left(|e_{low} - M_\epsilon|, |e_{high} - M_\epsilon|\right)}{IQR_\epsilon} \tag{27}$$

where e_{low} and e_{high} are the residual values at lower and upper quantiles.

3) Spectral consistency.

Spectral consistency evaluates the fidelity of key frequency components during denoising:

$$\delta = \left| \frac{A_{\hat{x}}(f_{main}) - A_x(f_{main})}{A_x(f_{main})} \right| \leq 100\% \tag{28}$$

where $A_x(f_{main})$ denotes the amplitude of the original signal at the main frequency and $A_{\hat{x}}(f_{main})$ represents the amplitude of the denoised signal at the main frequency.

Based on the above content regarding the AMFFS multi-modal fusion smoothing strategy, the parameter tuning of BOA-VMD and the configuration of AMFFS are divided into offline training and online inference

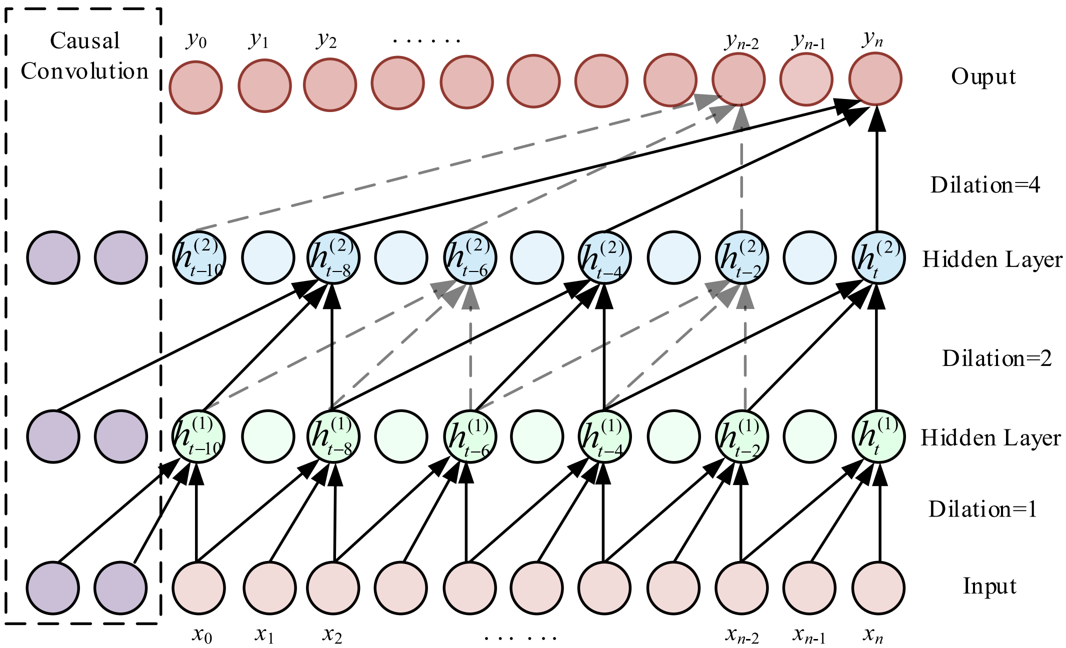


Fig. 4. Dilated convolution architecture diagram.

stages, as elaborated below. The BOA-VMD algorithm first determines the optimal decomposition layers k and penalty factor r through an iterative global optimization process during the offline training phase. This optimization process is computationally intensive and thus completed offline during the model training stage; once the optimal parameters are obtained, they are fixed for all subsequent decomposition tasks. In the online deployment stage on edge devices, such as the Raspberry Pi, the VMD decomposition directly applies these pre-optimized parameters without re-running the BOA. The decomposition of each newly incoming load sequence using a sliding window is then reduced to a non-iterative, deterministic signal processing operation, which is computationally lightweight and suitable for real-time or near-real-time rolling updates. The same principle applies to the AMFFS denoising step, which uses pre-computed entropy thresholds and fusion criteria derived offline. Therefore, while the parameter optimization is offline, the decomposition-denoising pipeline can still operate efficiently on edge devices for streaming load data, ensuring both high accuracy and low latency in practical deployment.

3.3. Short-term load forecasting model considering the comprehensive impacts of DR

After BOA-VMD decomposition and AMFFS-based denoising, the load forecasting task in the VPP scenario is transformed into a multivariate sequence modeling problem involving denoised load information, meteorological variables, and DR-related features. Although the previous stages effectively suppress random noise and preserve key load characteristics, the input sequence still exhibits strong non-stationarity, nonlinear fluctuations, multi-timescale periodicity, and delayed coupling effects among heterogeneous variables. Therefore, a single temporal modeling module is insufficient to simultaneously capture local fluctuation patterns, long-range sequential dependence, and the dynamic importance of different time steps. To address this issue, this paper constructs a BiTCN-BiGRU-Multi-Head Attention forecasting model. In this framework, BiTCN is employed to extract local and multi-scale temporal patterns, BiGRU is used to model sequential dependence and delayed contextual interactions, and the MHA is introduced to adaptively emphasize critical temporal features.

(1) Bidirectional temporal convolutional network.

In deep learning-based time-series forecasting, the architectural design of the temporal encoder plays a decisive role in capturing complex patterns and temporal dependencies. In this study, a bidirectional temporal convolutional network (BiTCN) is adopted as the front-end feature extractor. BiTCN combines the advantages of temporal convolutional networks and bidirectional sequence encoding, allowing the model to effectively capture short-term local fluctuations as well as medium- and long-term periodic structures in the input sequence. By stacking one-dimensional dilated convolutional layers, BiTCN can significantly enlarge the receptive field without excessively increasing the number of model parameters, thereby achieving efficient multi-scale temporal representation.

The convolution operation of the l-th forward branch can be expressed as:

$$\overrightarrow{h}_t^{(l)} = \sigma\left(\sum_{i=0}^{k-1} w_i^{(l,f)} x_{t-d_l i}^{(l-1)} + b^{(l,f)}\right) \tag{29}$$

and the backward branch is formulated as follows:

$$\overleftarrow{h}_t^{(l)} = \sigma\left(\sum_{i=0}^{k-1} w_i^{(l,b)} x_{t+d_l i}^{(l-1)} + b^{(l,b)}\right) \tag{30}$$

where $\overrightarrow{h}_t^{(l)}$ and $\overleftarrow{h}_t^{(l)}$ denote the forward and backward hidden representations at time step t, respectively; k is the convolution kernel size; d_l is the dilation factor of the l-th layer;$w_i^{(l,f)}$and$w_i^{(l,b)}$are the convolution kernel weights of the forward and backward branches, respectively;$b^{(l,f)}$ and$b^{(l,b)}$are the bias vectors; and $\sigma(\cdot)$ is the activation function.

To obtain the bidirectional temporal representation, the outputs of the two branches are concatenated as:

$$h_t^{(l)} = Concat\left(\overrightarrow{h}_t^{(l)}, \overleftarrow{h}_t^{(l)}\right) \tag{31}$$

In the proposed model, BiTCN is selected for three main reasons. First, compared with ordinary one-dimensional convolution, dilated convolution can expand the receptive field exponentially, which is beneficial for modeling day-level and week-level load periodicity without introducing excessive computational cost. Second, compared with purely recurrent structures, convolutional computation is more

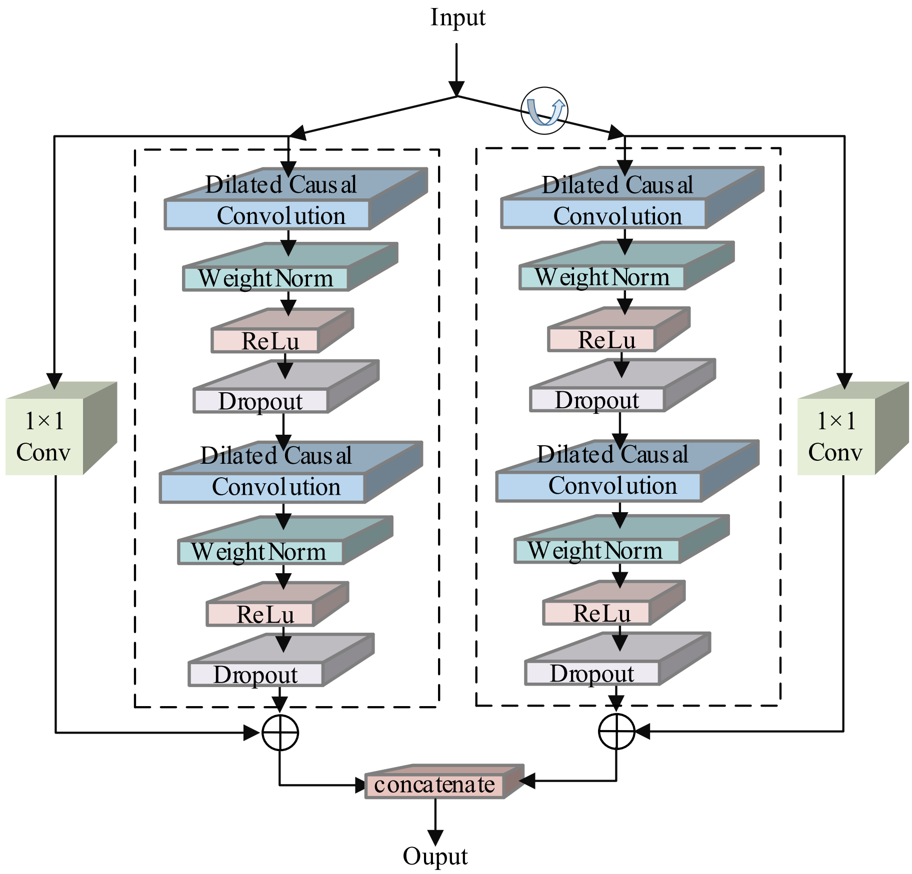


Fig. 5. Bidirectional temporal convolution structure.

parallelizable and therefore more suitable for efficient implementation, which is consistent with the edge deployment objective of this study. Third, the bidirectional structure allows the encoder to learn richer contextual information from both directions of the observed input window, thereby improving temporal representation quality. It should be noted that the bidirectional design is used to enhance the encoding of the historical observation window rather than to introduce unavailable future target information into forecasting. The BiTCN architecture is shown in Fig. 4.

In practical VPP load forecasting, these characteristics enable BiTCN to effectively handle load data with evident multi-timescale variation. For example, by capturing daily periodicity, the network can learn the load variation pattern within one day; by capturing weekly periodicity, it can recognize fluctuations across different weekdays; and by capturing

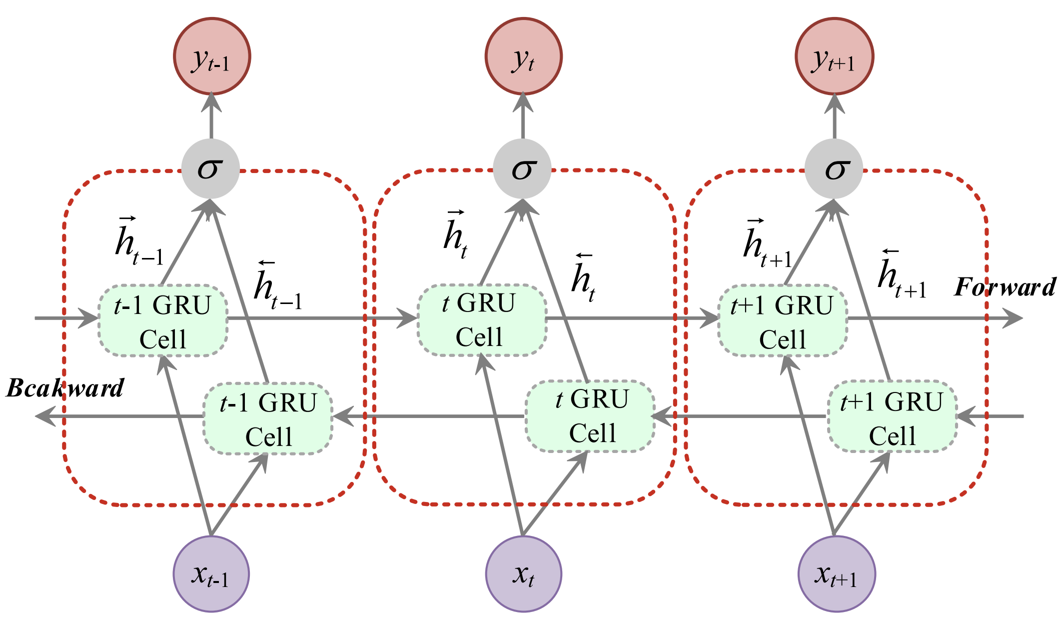


Fig. 6. BiGRU structure.

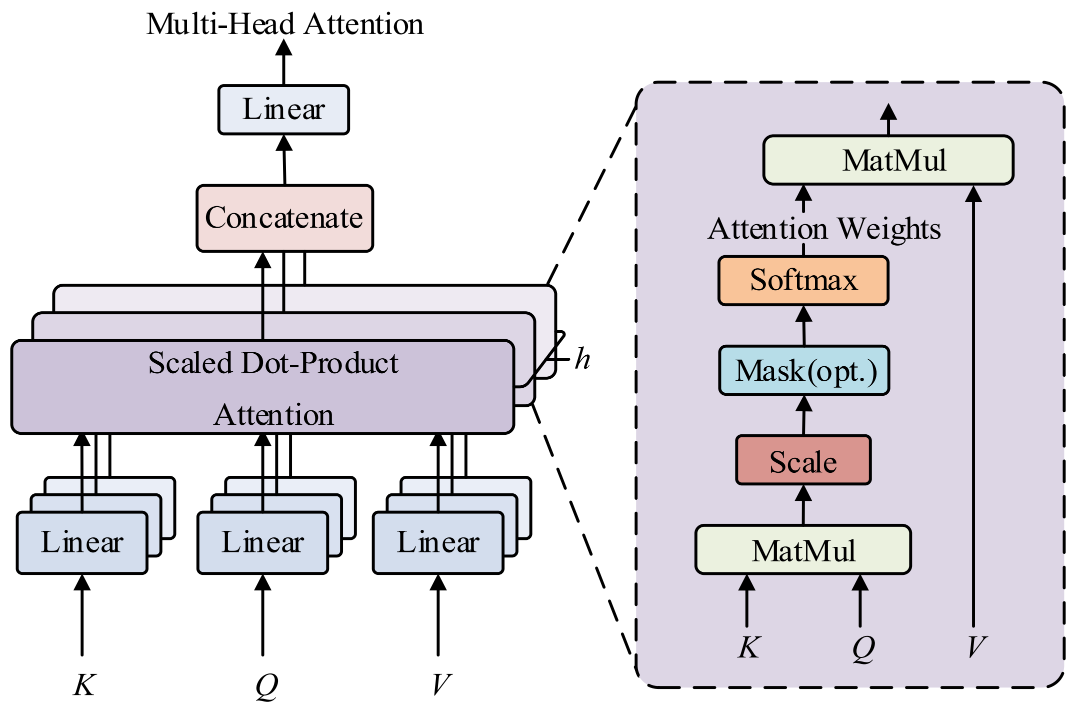


Fig. 7. Multi-head attention mechanism diagram.

seasonal changes, it can identify long-term variation trends under different environmental conditions. This multi-scale feature extraction capability enables BiTCN to provide a robust temporal representation basis for subsequent forecasting, as shown in Fig. 5.

(2) Bidirectional gated recurrent unit.

Although BiTCN is effective in extracting local and multi-scale temporal features, the load sequence still contains dynamic sequential dependence and delayed cumulative effects that require further modeling. To this end, a bidirectional gated recurrent unit (BiGRU) is introduced after the BiTCN module. BiGRU consists of a forward GRU and a backward GRU. The forward GRU processes the sequence in chronological order and captures the influence of previous observations on the current hidden state, whereas the backward GRU processes the sequence in reverse order and provides complementary contextual information within the observed input window. By combining the hidden states of the two branches, BiGRU generates a more comprehensive representation of temporal dynamics. The BiGRU architecture is shown in Fig. 6.

Furthermore, the gating mechanism of GRU effectively alleviates the gradient vanishing problem found in traditional recurrent neural networks, allowing the model to better handle long time-series data. The core computation process of GRU includes three key components: the reset gate, update gate, candidate hidden state, and final hidden state. The mathematical expressions are as follows:

Reset gate:

$$R_t = \sigma(W_{xr}x_t + W_{hr}h_{t-1} + b_r) \tag{32}$$

Update gate:

$$Z_t = \sigma(W_{xz}x_t + W_{hz}h_{t-1} + b_z) \tag{33}$$

Candidate hidden state:

$$\widetilde{H}_t = tanh(x_t W_{xh} + (r_t \odot h_{t-1})W_{hh} + b_h) \tag{34}$$

Final hidden state:

$$H_t = z_t \odot h_{t-1} + (1 - z_t) \odot \widetilde{h}_t \tag{35}$$

where x_t is the input feature vector at time step t, $h_{t\text{-}1}$ represents the previous hidden state, $W_{(\cdot)}$ denotes the weight matrix; $b_{(\cdot)}$ is the bias vector, $\sigma(\cdot)$ is the sigmoid activation function, $\tanh(\cdot)$ is the hyperbolic tangent activation function, and $\odot$ denotes element-wise multiplication.

The reason for selecting BiGRU rather than more complex recurrent units is threefold. First, compared with BiLSTM, BiGRU has a simpler gating structure and fewer trainable parameters, which reduces training complexity and computational burden. This is particularly suitable for this study, because the forecasting model is required not only to achieve high accuracy but also to support edge deployment on Raspberry Pi devices. Second, after BiTCN has already extracted rich local and multi-scale features, the recurrent module mainly needs to further characterize sequential dependence and delayed response effects. Under this setting, BiGRU offers a favorable balance between expressive capability and computational efficiency. Third, the gating mechanism of GRU is well suited to modeling delayed and accumulated impacts in DR scenarios, since user responses to price signals often do not occur instantaneously but evolve gradually over time.

In practical VPP load forecasting, the role of BiGRU is not limited to preserving temporal order. More importantly, it learns the dynamic interaction among denoised load features, meteorological variables, and DR-related information. For example, a sudden change in electricity price may trigger a delayed user response, and temperature or humidity may influence load demand with a temporal lag. BiGRU can effectively capture such nonlinear and lagged dependencies, thereby enhancing the model's ability to characterize demand-side dynamics.

(3) Multi-head attention mechanism.

Although BiTCN and BiGRU can jointly extract rich temporal representations, not all hidden features contribute equally to the final load forecasting result. In the VPP context, the importance of input information varies significantly across time steps and operating conditions. For example, during peak-load periods, certain historical observations may be more informative than others; during extreme weather events, meteorological variables may play a dominant role; and during periods with strong price-responsive user behavior, behavioral features may exert a stronger influence on load variation. Therefore, an MHA is introduced to adaptively assign different importance weights to temporal features.

The MHA is an extension of the self-attention mechanism. By computing multiple attention distributions in parallel in different subspaces, it enables the model to capture diverse dependency patterns from multiple perspectives. In short-term power load forecasting, the MHA can dynamically reweight the feature sequence generated by BiGRU, thereby highlighting those time steps and hidden features that are most relevant to the forecasting target. For instance, specific periods such as holidays, abrupt electricity price changes, extreme weather events, or intensified user-side behavioral adjustments may have a disproportionately large impact on future load. Through adaptive weighting, the attention module allows the model to focus more effectively on these critical temporal regions. This not only improves forecasting accuracy, but also enhances the model's ability to characterize nonlinear periodicity, temporal heterogeneity, and multi-scale variation in load data. The architecture of the MHA is shown in Fig. 7.

The MHA can be divided into single-head attention and multi-head attention. Single-head attention calculates the association weights of different time steps in the input sequence through the interaction of the query (Q), key (K), and value (V) matrices. Its computation formula is shown in Eq. (36):

$$\mathrm{Attention}(Q,K,V) = \mathrm{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V \tag{36}$$

where$Q \in \mathbb{R}^{T\times d_k}$,$K \in \mathbb{R}^{T\times d_k}andV \in \mathbb{R}^{T\times d_k}$ are the query, key, and value matrices, T is the number of time steps, and d_k is the dimensionality of the key vector.

Multi-head attention performs parallel computations of attention distributions in h independent subspaces, enhancing the model's ability to capture different patterns. Its computation formula is shown in Eq. (37):

$$\begin{cases} MultiHead(Q,K,V) = Concat(head_1,\dots,head_h)W^O \\ \qquad head_i = Attention\left(QW_i^Q, KW_i^K, VW_i^V\right) \end{cases} \tag{37}$$

where W_i^Q, W_i^K and W_i^V are the projection matrices of the i-th attention head, W^O is the output weight matrix.

Compared with directly using recurrent hidden states for prediction, the MHA has two key advantages. First, it can explicitly distinguish the relative importance of different temporal positions, thereby enabling the model to focus on the most informative historical features. Second, multi-head attention performs parallel learning in multiple representation subspaces, which allows different heads to capture different aspects of the sequence, such as local fluctuations, periodic structures, price-responsive load variations, and weather-sensitive patterns. This is particularly suitable for VPP load forecasting, where the load formation mechanism is jointly affected by multiple heterogeneous factors.

In summary, the proposed BiTCN-BiGRU-Multi-Head Attention architecture is not a simple stacking of deep learning modules. Instead, it is a targeted design for VPP short-term load forecasting under DR conditions. BiTCN is used to efficiently extract multi-scale local temporal patterns, BiGRU is employed to model sequential dependence and delayed response effects, and the MHA is introduced to adaptively emphasize critical information among heterogeneous features. Through the coordinated operation of these modules, the proposed model can more effectively capture the non-stationarity, multi-timescale variation, delayed user response, and strong coupling characteristics of VPP load data.

(4) DR mechanism based on logistic function.

In traditional short-term load forecasting for power systems, forecasting models often neglect the influence of user response behaviors, with input features primarily relying on external environmental factors such as temperature, weather conditions, and day types. The composition of input features in traditional neural network models can be expressed as Eq. (38):

$$\begin{pmatrix} L_{day1}^{time1} & L_{day1}^{time2} & \cdots & L_{day1}^{time95} & L_{day1}^{time96} & T_{day1}^{max} & T_{day1}^{ave} & T_{day1}^{min} & H_{day1}^{ave} & R_{day1}^{mm} \\ L_{day2}^{time1} & L_{day2}^{time2} & \cdots & L_{day2}^{time95} & L_{day2}^{time96} & T_{day2}^{max} & T_{day2}^{ave} & T_{day2}^{min} & H_{day2}^{ave} & R_{day2}^{mm} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ L_{dayk-1}^{time1} & L_{dayk-1}^{time2} & \cdots & L_{dayk-1}^{time95} & L_{dayk-1}^{time96} & T_{dayk-1}^{max} & T_{dayk-1}^{ave} & T_{dayk-1}^{min} & H_{dayk-1}^{ave} & R_{dayk-1}^{mm} \\ L_{dayk}^{time1} & L_{dayk}^{time1} & \cdots & L_{dayk}^{time95} & L_{dayk}^{time96} & T_{dayk}^{max} & T_{dayk}^{ave} & T_{dayk}^{ave} & H_{dayk}^{ave} & R_{dayk}^{mm} \end{pmatrix} \tag{38}$$

where *Ltimei dayk* represents the historical load data at the i-th hour of the k-th day; T_{dayk}^{max}, T_{dayk}^{ave}, and T_{dayk}^{min} represent the maximum, average, and minimum temperatures of the k-th day, respectively; H_{dayk}^{ave} represents the humidity of the k-th day; and *Rmm dayk* represents the precipitation of the k-th day.

However, with the widespread implementation of DR mechanisms, whether through price-based or incentive-based policies, the user load curve changes due to DR, showing a trend of peak shaving and valley filling. This nonlinear variation renders the traditional input features unsuitable, making it difficult to accurately model the load evolution patterns. In price-based incentive scenarios, user response behavior is influenced by factors such as price differences, price change gradients, and the smoothness of the response curve, exhibiting significant randomness. The actual DR curve lies between optimistic response and pessimistic response, presenting fuzzy uncertainty. To characterize this bounded, gradual, and saturating response behavior in a mathematically consistent and behaviorally realistic manner, the selection of a suitable response function becomes critical.

Several types of functions are commonly used to model price-responsive user behavior. Linear functions, although simple, cannot capture the inertial region feature or response-limited constraints observed in actual DR scenarios. Exponential functions can represent rapid growth but lack an upper bound, contradicting the physical limits of load adjustability, and they over-emphasize marginal changes at small price differences. Step functions with hard thresholds ignore the gradual cognitive transition that users exhibit when price differences increase, leading to unrealistic abrupt changes in response intensity. Other sigmoidal alternatives, such as the Gompertz function or the arctangent function, also exhibit S-shaped characteristics. However, the

Table 2
Behavioral economics interpretation of DRM-TD model parameters.

Parameter	Behavioral economics principle	Interpretation
ν	Bounded rationality	Feasible range of load adjustment, constrained by the size of flexible load resources.
δ	Anchoring effect	Psychological reference point where users begin to perceive price incentives as economically meaningful.
μ	Behavioral transition smoothness	Controls how fast users transition from old habits to a new equilibrium; smaller μ means steeper curve.
η	Marginal utility / asymmetry	Adjusts the response baseline; positive η indicates a more active response tendency under the same price signal.
κ (in Eq. (40))	Fuzzy preference tendency	Determines the relative position of the actual response trajectory between the optimistic and pessimistic response boundaries.
ρ (in Eq. (41))	Behavioral inertia	Time constant for adapting to price signals; smaller ρ means faster response.

Gompertz function is asymmetric and tends to saturate more slowly on one side, which does not align well with the approximately symmetric responsiveness of users to price increases and decreases in the absence of strong behavioral biases. The arctangent function, while symmetric, approaches the response-limited state at a much slower rate, making it less sensitive to price differences in the elastic response region and thereby weakening its capability to identify mid-range behavioral transitions.

Based on the above analysis, this paper proposes a logistic-function-driven delay-response model (DRM-TD), which introduces user response randomness and optimistic response membership as constraints to achieve precise modeling of fuzzy behavioral attributes. The logistic function, a special form of the sigmoid family, inherently models the three typical phases of user behavior: an initial inertial region where users are insensitive to very small price differences, a progressive response region where the response increases in an approximately linear but sublinear manner, and a response-limited region where the response reaches its ceiling. This S-shaped characteristic aligns naturally with consumer psychology: users do not react immediately to tiny price changes, become increasingly sensitive as the incentive grows, and eventually reach their adjustment limit. Moreover, the logistic function is continuously differentiable, ensuring smooth gradient propagation when integrated into deep learning frameworks. Its parameters, namely ν for response range, δ for sensitivity center, μ for transition steepness, and η for baseline asymmetry, have clear physical interpretations that can be calibrated from historical load and price data. Compared to the Gompertz or arctangent functions, the logistic function provides a symmetric response curve around δ, which better reflects the realistic property that, without behavioral biases, users' sensitivity to price increases and decreases is approximately symmetric. Therefore, the logistic function offers an optimal trade-off between behavioral fidelity, mathematical tractability, and parameter identifiability, making it a well-justified choice for the proposed DRM-TD model.

From a behavioral economics perspective, user response to electricity price differences is bounded, threshold-dependent, and gradual rather than instantaneous and perfectly rational. When the price difference is very small, the perceived economic benefit is generally insufficient to overcome users' habitual consumption patterns, adjustment costs, or comfort constraints. As the price difference increases, users gradually recognize the economic incentive and begin to adjust their electricity consumption behavior. Once the price gap grows sufficiently large, the adjustable load potential is nearly depleted, and the response gradually enters a response-limited stage. Accordingly, the Logistic function is adopted to naturally characterize this threshold-triggered elastic-to-restricted response evolution. Meanwhile, optimistic and pessimistic response boundaries, together with the corresponding membership degrees, are further introduced to depict the fuzzy uncertainty and heterogeneity inherent in practical user behaviors. The logistic function is expressed as Eq. (39):

$$\psi_{p-v,t}(\Delta\omega_{p-v,t}) = \frac{\nu}{1+e^{-(\Delta\omega_{p-v,t}-\delta)/\mu}} + \eta \tag{39}$$

where ν is the response span coefficient, representing the feasible range of user-side load adjustment; δ is the response sensitivity center, which determines the midpoint of the progressive response region; μ is the shape coefficient controlling the smoothness of the transition process; and η is the asymmetry parameter that adjusts the response baseline under different behavioral tendencies. Table 2 summarizes the behavioral economics interpretation of each parameter.

To further represent heterogeneity, the optimistic response curve and pessimistic response curve are introduced as two bounding scenarios. The former represents an active load-adjustment tendency, while the latter corresponds to a conservative or inertia-dominated tendency. The optimistic-response membership degree μ determines the relative position of the actual response trajectory between these two bounds. Thus, the model does not assume identical user behavior; instead, it characterizes the actual DR trajectory as a fuzzy outcome bounded by optimistic and pessimistic limits. In addition, the delay factor accounts for the temporal lag between price signal release and actual load adjustment, since user-side decisions and device-level execution cannot be instantaneous. Specifically:

When $\Delta\omega_{p\text{-}v,t} \le m_{p\text{-}v,t}$, the DR behavior enters a randomly dominated state within the inertial region. In this region, price signals exert only a weak influence on user decision-making, and DR is primarily governed by personal preferences and random disturbances. Accordingly, an accurate DR model can be established by adopting the weighted average of the optimistic and pessimistic response estimation curves.

When $n_{p\text{-}v,t} \le \Delta\omega_{p\text{-}v,t}$, the user's response potential reaches a fully saturated state. In this phase, the optimistic response curve and the pessimistic response curve overlap, and the precise DR can be directly determined.

When $m_{p\text{-}v,t} \le \Delta\omega_{p\text{-}v,t} \le n_{p\text{-}v,t}$, the user exhibits price elasticity behavior, where their responsiveness increases gradually with the $\Delta\omega_{p\text{-}v,t}$. In this range, the DR trajectory follows a monotonic asymptotic path toward the optimistic curve. Accordingly, the response behavior under various stages can be expressed as Eqs. (40) and (41):

$$\widehat{\psi}_{p-v,t} = \begin{cases} \dfrac{\psi_{max,p-v,t}+\psi_{min,p-v,t}}{2}, & 0 \le \Delta\omega_{p-v,t} \le m_{p-v,t} \\ \psi_{min,p-v,t} + \dfrac{\psi_{max,p-v,t}-\psi_{min,p-v,t}}{2}(1+\kappa), & m_{p-v,t} \le \Delta\omega_{p-v,t} \le n_{p-v,t} \\ \psi_{max,p-v,t}, & n_{p-v,t} \le \Delta\omega_{p-v,t} \end{cases} \tag{40}$$

$$\widetilde{\psi}_{p-v,t}(t) = \rho\widetilde{\psi}_{p-v,t}(t-1) + (1-\rho)\widehat{\psi}_{p-v,t}(t) \tag{41}$$

where $\widehat{\psi}_{p-v,t}$ denotes the baseline DR rate without delay effects, and $\widetilde{\psi}_{p-v,t}$ is the modified DR incorporating temporal delay characteristics.

Eq. (41) defines the instantaneous user response under the DR model with time-delay effects. The parameters $\psi_{max,p-v,t}$ and $\psi_{min,p-v,t}$ respectively is the maximum and minimum DR thresholds under optimistic and pessimistic scenarios. The optimistic response index κ can be calculated as shown in Eq. (42), and the delay factor ρ can be calculated as shown in Eq. (43).

$$\kappa = \frac{\Delta\omega_{p-v,t} - m_{p-v,t}}{n_{p-v,t} - m_{p-v,t}}, \qquad m_{p-v,t} \le \Delta\omega_{p-v,t} \le n_{p-v,t} \tag{42}$$

$$\rho = e^{-\Delta t/\tau} \tag{43}$$

However, for practical load forecasting and DR evaluation, the instantaneous response at each sampling instant must be further aggregated into the net load shifting quantity over each pricing period.

In this paper, Q_t denotes the net load variation from DR in period t. Its formulation adheres to load transfer conservation: incoming load is positive, outgoing negative. Under this rule, the period-wise net transfer can be derived by examining inter-period flows among peak, flat, and valley periods.

During peak periods ($t \in t_p$), the system acts as a net load exporting interval driven by price incentives. As no additional load is shifted into these time slots, the net load shifting amount is solely determined by the load diverted from peak hours to flat and valley periods:

$$Q_t = -\widetilde{\omega}_{p-f,t}\widetilde{\epsilon}_p - \widetilde{\omega}_{p-v,t}\widetilde{\epsilon}_p \tag{44}$$

During the flat period ($t \in t_f$), both load inflow and outflow coexist. Specifically, this period receives part of the shifted load from the peak period, while a portion of its native load is further transferred to the valley period. Accordingly, the net load shifting magnitude for the flat period is defined as the difference between the incoming transferred load and the outgoing transferred load:

$$Q_t = \widetilde{\omega}_{p-f,t}\widetilde{\epsilon}_p - \widetilde{\omega}_{f-v,t}\widetilde{\epsilon}_f \tag{45}$$

During the valley period ($t \in t_v$), this interval acts as a net load-importing region. Since no load is diverted out of the valley period, its net load shifting magnitude equals the total load transferred into this interval from both the peak and flat periods:

$$Q_t = \widetilde{\omega}_{p-v,t}\widetilde{\epsilon}_v + \widetilde{\omega}_{f-v,t}\widetilde{\epsilon}_f, \tag{46}$$

Based on the above method, the DR rates for the transitions from peak to flat and from flat to valley periods can be mathematically derived. Therefore, the load transfer generated by actual DR is expressed as Eq. (47):

$$Q_t = \begin{cases} -\widetilde{\omega}_{p-f,t}\widetilde{\epsilon}_p - \widetilde{\omega}_{p-v,t}\widetilde{\epsilon}_p, & t \in t_p \\ \widetilde{\omega}_{p-f,t}\widetilde{\epsilon}_p - \widetilde{\omega}_{f-v,t}\widetilde{\epsilon}_f, & t \in t_f \\ \widetilde{\omega}_{p-v,t}\widetilde{\epsilon}_v + \widetilde{\omega}_{f-v,t}\widetilde{\epsilon}_f, & t \in t_v \end{cases} \tag{47}$$

where t_p, t_f, and t_v respectively denote the peak, flat, and valley time periods;$\widetilde{\epsilon}_p$,$\widetilde{\epsilon}_f$, and$\widetilde{\epsilon}_v$represent the corresponding average load levels during these periods.

Consequently, the load transfer quantity derived from the proposed DRM-TD model serves as a critical behavioral feature that reflects price-driven user adjustments. By embedding this feature into the input structure, the forecasting model can effectively capture the dynamic coupling between electricity prices and load variations.

$$\begin{pmatrix} L_{day1}^{time1}+S_{day1}^{DR1} & L_{day1}^{time2}+S_{day1}^{DR2} & \cdots & L_{day1}^{time95}+S_{day1}^{DR95} & L_{day1}^{time96}+S_{day1}^{DR96} & T_{day1}^{max} & T_{day1}^{ave} & T_{day1}^{min} & H_{day1}^{ave} & R_{day1}^{mm} \\ L_{day2}^{time1}+S_{day2}^{DR1} & L_{day2}^{time2}+S_{day2}^{DR2} & \cdots & L_{day2}^{time95}+S_{day2}^{DR95} & L_{day2}^{time96}+S_{day2}^{DR96} & T_{day2}^{max} & T_{day2}^{ave} & T_{day2}^{min} & H_{day2}^{ave} & R_{day2}^{mm} \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ L_{dayk-1}^{time1}+S_{dayk-1}^{DR1} & L_{dayk-1}^{time2}+S_{dayk-1}^{DR2} & \cdots & L_{dayk-1}^{time95}+S_{dayk-1}^{DR95} & L_{dayk-1}^{time96}+S_{dayk-1}^{DR96} & T_{dayk-1}^{max} & T_{dayk-1}^{ave} & T_{dayk-1}^{min} & H_{dayk-1}^{ave} & R_{dayk-1}^{mm} \\ L_{dayk}^{time1}+S_{dayk}^{DR1} & L_{dayk}^{time1}+S_{dayk}^{DR2} & \cdots & L_{dayk}^{time95}+S_{dayk}^{DR95} & L_{dayk}^{time96}+S_{dayk}^{DR96} & T_{dayk}^{max} & T_{dayk}^{ave} & T_{dayk}^{ave} & H_{dayk}^{ave} & R_{dayk}^{mm} \end{pmatrix} \tag{48}$$

where *SDRi dayk* denotes the load transfer quantity at hour *i* on day *k*.

(5) Sensitivity analysis of key DRM-TD parameters.

From the perspective of practical model identification, the parameters in the DRM-TD model are calibrated at the aggregated load level using real historical load and price data from Fujian Province, rather than being identified from individual user-level behavioral observations. Accordingly, the parameter set is selected with reference to the observed price-driven peak-valley shifting characteristics in the regional dataset, and should therefore be interpreted as a group-level behavioral descriptor of the target user population rather than a set of universal constants directly applicable to all user types. In this sense, ν, δ, μ, η, κ, and ρ jointly characterize the aggregate response capability, behavioral hesitation, fuzzy preference tendency, and delayed adjustment characteristics of the user group under study.

To guarantee the analytical rigor and rationality of parameter calibration, a mathematical sensitivity analysis is performed for the core parameters of the DRM-TD model. This analysis quantifies the distinct impacts of each parameter on user response characteristics across the inertial region, elastic response region, and response-limited region, respectively. The parameter configurations adopted in the analysis are presented in Table 3.

1) Sensitivity analysis of the response sensitivity center δ.

By taking the partial derivative of the logistic response function with respect to δ, we obtain:

$$\frac{\partial \psi_{p-v,t}}{\partial \delta} = -\frac{\nu e^{-\left(\Delta\gamma_{p-v,t}-\delta\right)/\mu}}{\mu\left(1+e^{-\left(\Delta\gamma_{p-v,t}-\delta\right)/\mu}\right)^2} \tag{49}$$

Inertial region: when $\Delta\omega_{p-v,t} \ll \delta, e^{-\left(\Delta\omega_{p-v,t}-\delta\right)/\mu} \rightarrow +\infty$, and$\frac{\partial \psi_{p-v,t}}{\partial \delta} \rightarrow 0$. δ has a negligible influence on the demand response rate within the inertial region, where user behavior is governed by randomness and remains largely unresponsive to price signals.

Response-limited region: when $\Delta\omega_{p-v,t} \gg \delta$, $e^{-\left(\Delta\omega_{p-v,t}-\delta\right)/\mu} \rightarrow 0$, and $\frac{\partial \psi_{p-v,t}}{\partial \delta} \rightarrow 0$. This indicates that user responsiveness has already reached its upper limit, and variations in δ exert negligible effects on the saturated response state.

Elastic response region: let$x = \Delta\omega_{p-v,t} - \delta$, and define $y = \frac{\partial \psi_{p-v,t}}{\partial \delta} = -\frac{\nu e^{-x/\mu}}{\mu(1+e^{-x/\mu})^2}$.$y'$with respect to δ can be analyzed through Eq. (50):

$$y' = \frac{\nu}{\mu^2} \cdot \frac{e^{-x/\mu}\left(1-e^{-x/\mu}\right)}{\left(1+e^{-x/\mu}\right)^3} \tag{50}$$

Table 3
The core parameters of the DRM-TD model.

Parameter	Optimistic value	Pessimistic value
δ	0.5	0.5
ν	0.1	0.104
μ	0.1	0.1
η	0	−0.003

From Eq. (50), sensitivity decreases monotonically for $x < 0$, $y' < 0$, and increases for $x > 0$, $y' > 0$. With $v > 0$, $\mu > 0$, the sensitivity function $y = \frac{\partial \psi_{p-v,t}}{\partial \delta}$ reaches a local minimum at $x = 0$, i.e., $\Delta\omega_{p-v,t} = \delta$, where $y|_{x=0} = -\frac{v}{4\mu} < 0$. Hence, δ is negatively correlated with the estimated response intensity $\psi_{p-v,t}$: a leftward shift of δ expands the hesitation region and overestimates responsiveness, while a rightward shift pushes part of the effective response region into the inertial zone, underestimating responsiveness. This finding is consistent with the anchoring effect in behavioral economics, where δ acts as the psychological reference point that determines users' sensitivity threshold to price incentives.

From a practical perspective, accurate calibration of δ is critical to model performance. An excessively small δ will overestimate user response ability and may cause the model to classify too many users as responsive, thereby reducing the response curve's peak-valley difference. An excessively large δ underestimates effective responses and enlarges the predicted load profile's peak-valley difference. Thus, δ should be calibrated near the midpoint of the effective response interval to avoid systematic overestimation or underestimation of user response capability.

2) Sensitivity analysis of response span v.

The partial derivative with respect to v is given by Eq. (51):

$$\frac{\partial \psi_{p-v,t}}{\partial v} = \frac{1}{1 + e^{-(\Delta\omega_{p-v,t} - \delta)/\mu}} \tag{51}$$

Eq. (51) shows how sensitive the function is to v, directly affected by the deviation between the $\Delta\omega_{p-v,t}$ and the response sensitivity center. Because the exponential term is always positive, the derivative remains within (0,1), implying that the effect of v on the response function is always positive but region-dependent.

Inertial region: Eq. (51) reveals that the sensitivity to v is influenced by $\Delta\omega_{p-v,t} - \delta$. Since $e^{-(\Delta\omega_{p-v,t}-\delta)/\mu} > 0$, it follows that $1 + e^{-(\Delta\omega_{p-v,t}-\delta)/\mu} > 1$. when $\Delta\omega_{p-v,t} \ll \delta$, DR behavior enters the inertial region. Define $k = \delta - \Delta\omega_{p-v,t}$, then $e^{-(\Delta\omega_{p-v,t}-\delta)/\mu} = e^{k/\mu} \to +\infty$. Substituting into the derivative yields:

$$\frac{\partial \psi_{p-v,t}}{\partial v} = \frac{1}{1 + e^{k/\mu}} \to 0 \tag{52}$$

Hence, when $\Delta\omega_{p-v,t}$ is extremely small, variations in the response span coefficient v have a negligible effect on $\psi_{p-v,t}(\Delta\omega_{p-v,t})$. In this regime, the model output can be considered practically insensitive to v, which aligns with the behavioral expectation that users do not respond to price signals within the inertial region.

Response-limited region analysis: when $\Delta\omega_{p-v,t} \gg \delta$, the DR behavior enters the response-limited region. At this time, $e^{-(\Delta\omega_{p-v,t}-\delta)/\mu} \to 0$, and $\lim_{\Delta\gamma_{p-v,t} \to +\infty} \frac{\partial \psi_{p-v,t}}{\partial v} \to 1$. This indicates that, in the response-limited region, changes in v almost proportionally affect the DR rate $\psi_{p-v,t}(\Delta\omega_{p-v,t})$. Thus, v directly determines the upper bound of response potential once users' transferable load resources are nearly fully activated.

Elastic response region: when $\Delta\omega_{p-v,t}$ lies within the elastic response region, let $x = \Delta\omega_{p-v,t} - \delta$, and define $y = \frac{\partial \psi_{p-v,t}}{\partial v} = \frac{1}{1+e^{-(\Delta\omega_{p-v,t}-\delta)/\mu}}$. y' with respect to x can be analyzed through Eq. (53):

$$y' = \frac{e^{-x/\mu}/\mu}{(1 + e^{-x/\mu})^2} > 0 \tag{53}$$

Eq. (53) further indicates that the sensitivity of the function with respect to ν increases monotonically with the transformed price difference $x = \Delta\omega_{p-v,t} - \delta$. This sensitivity exhibits an initial rapid growth phase, followed by a gradual deceleration.

Specifically, when $x < 0$ (i.e., $\Delta\omega_{p-v,t} < \delta$), the sensitivity curve presents a convex feature, with its rising speed continuing to accelerate. When $x > 0$ (i.e., $\Delta\omega_{p-v,t} > \delta$), the curve turns concave, and the growth trend slows down gradually. At the inflection point $x = 0$ (i.e., $\Delta\omega_{p-v,t} = \delta$), the derivative of the sensitivity reaches the peak value:

$$y' = \frac{1}{4\mu} \tag{54}$$

This result demonstrates that the sensitivity of the response to the span coefficient v follows an S-shaped asymptotic pattern across the progressive response interval, where the inflection point corresponds to the maximum growth rate of sensitivity. The foregoing analysis leads to the following conclusions regarding the role of the response span coefficient v in different operating regimes.

In the response-limited region, the model output exhibits an approximately proportional sensitivity to variations in v. Specifically, an underestimation of v results in an undervaluation of the maximum response capacity of users, thereby causing the model to understate the peak-shaving potential. Consequently, the resulting response curve maintains a pronounced peak-to-valley disparity. Conversely, overestimating the parameter v inflates the perceived user responsiveness, which yields an overly smoothed peak-valley deviation on the response profile. Within the inertial interval, v exerts a negligible impact on system response, meaning parametric errors in this region barely interfere with model forecasting accuracy. By contrast, in the elastic response range, variations in v strongly regulate the simulated outputs. Precise calibration of v is essential to guarantee model reliability during the transition of user behavioral trends, and this calibration further governs the rational configuration of incentive strengths in DR schemes. Improper specification of v across the elastic response domain may either push the incentive mechanism into an early response-saturated state or fail to mobilize adequate user participation in demand response activities.

3) Sensitivity analysis of the shape coefficient μ.

Eq. (55) gives the partial derivative of the response function relative to μ:

$$\frac{\partial \psi_{p-v,t}}{\partial \mu} = -\frac{v(\Delta\omega_{p-v,t} - \delta)e^{-(\Delta\omega_{p-v,t}-\delta)/\mu}}{\mu^2\left(1 + e^{-(\Delta\omega_{p-v,t}-\delta)/\mu}\right)^2} \tag{55}$$

Analysis of Eq. (55) reveals the following behavior:

At the sensitivity center: When the electricity price difference equals the sensitivity center, i.e., $\Delta\omega_{p-v,t} = \delta$, the numerator contains the factor $(\Delta\omega_{p-v,t} - \delta) = 0$. Consequently,

$$\frac{\partial \psi_{p-v,t}}{\partial \mu} = 0 \tag{56}$$

Under this condition, variations in μ have no effect on $\psi_{p-v,t}(\Delta\omega_{p-v,t})$. When $\Delta\omega_{p-v,t}$ deviates from δ, the sensitivity exhibits nonlinear changes.

Inertial region: when the price difference is sufficiently small such that $\Delta\omega_{p-v,t} \ll \delta$, Let $k = \delta - \Delta\omega_{p-v,t}$, the user response enters the inertial region. Define $k = \delta - \Delta\omega_{p-v,t} > 0$. Then:

$$e^{-(\Delta\omega_{p-v,t}-\delta)/\mu} = e^{k/\mu} \to +\infty \tag{57}$$

Since $e^{k/\mu} \gg 1$, the denominator can be approximated as:

$$\mu^2(1 + e^{k/\mu})^2 \approx \mu^2 e^{2k/\mu} \tag{58}$$

Substituting into the derivative yields:

$$\frac{\partial \psi_{p-v,t}}{\partial \mu} \approx -\frac{-vke^{k/\mu}}{\mu^2 e^{2k/\mu}} = \frac{vk}{\mu^2 e^{k/\mu}} \to 0 \tag{59}$$

Thus, when the price difference is much smaller than the sensitivity center, the user response remains in the inertial region, and users exhibit no reaction to the small price differential. Variations in the shape coefficient μ cannot alter the inertial-region-dominated state, and the corresponding partial derivative asymptotically approaches zero. In this regime, the response function is effectively insensitive to μ.

Response-limited region: when$\Delta\omega_{p-v,t}$ is substantially larger than the sensitivity center, i.e., $\Delta\omega_{p-v,t} \gg \delta$, the exponential term satisfies$e^{-(\Delta\omega_{p-v,t}-\delta)/\mu} \to 0$, Consequently, the denominator $\mu^2\left(1+e^{-(\Delta\omega-\delta)/\mu}\right)^2 \to \mu^2$. Substituting into the partial derivative yields:

$$\frac{\partial\psi_{p-v,t}}{\partial\mu} \to 0 \tag{60}$$

Eq. (60) indicates that when$\Delta\omega_{p-v,t}$lies deep in the response-limited region, the user's DR potential is exhausted. Variations in the shape coefficient μ cannot alter the saturated response state, while the corresponding partial derivative gradually approaches zero. In this regime, the response function is effectively insensitive to μ.

Elastic response region: defined$\Delta\omega_{p-v,t} - \delta$. The corresponding partial differential with respect to μ may be expressed alternatively as:

$$\frac{\partial\psi_{p-v,t}}{\partial\mu} = -\frac{vxe^{-x/\mu}}{\mu^2\left(1+e^{-x/\mu}\right)^2} \tag{61}$$

By differentiating this relation with respect to x, we obtain:

$$\frac{\partial^2\psi_{p-v,t}}{\partial\mu\partial x} = -\frac{\nu e^{-x/\mu}}{\mu^2\left(1+e^{-x/\mu}\right)^3}\left[\left(1+e^{-x/\mu}\right) - \frac{x\left(1-e^{-x/\mu}\right)}{\mu}\right] \tag{62}$$

An analysis of Eq. (62) reveals the inherent characteristics within the elastic response interval: when$x < 0$(i.e.,$\Delta\omega_{p-v,t} < \delta$), the relevant partial differential$\frac{\partial\psi_{p-v,t}}{\partial\mu}$declines with the growth of x. When$x > 0$(i.e.,$\Delta\omega_{p-v,t} > \delta$), this differential term presents a trend of initially falling and then rising as x increases. These findings demonstrate that the varying rate of user responsiveness is dominated by the parameter μ. A smaller value of μ accelerates the dynamic change in response intensity, rendering user behavioral responses more susceptible to minor fluctuations in electricity price differentials. An excessively small μ will narrow the peak-to-valley disparity of the response profile. In contrast, a larger μ slows down the shifting rate of the response level and introduces a more noticeable lag in users' behavioral adjustments. If μ is set to an overly high value, the peak-to-valley deviation of the resulting response curve will become more prominent.

4) Sensitivity analysis of the asymmetry parameter η.

Eq. (63) gives the partial derivative of the response function relative to η:

$$\frac{\partial\psi_{p-v,t}}{\partial\eta} = 1 \tag{63}$$

Eq. (63) indicates that$\psi_{p-v,t}\left(\Delta\omega_{p-v,t}\right)$varies linearly with η under all conditions. Irrespective of the settings of other structural parameters, adjustments to η exert a proportional influence on$\psi_{p-v,t}\left(\Delta\omega_{p-v,t}\right)$. Accordingly, the responsiveness of the function toward η remains constant throughout the entire interval.

To render the response function insensitive to this parameter, the theoretical optimal value of η equals zero. Nevertheless, assigning $\eta = 0$ introduces an implicit premise that users hold symmetric perceptual sensitivity toward both positive and negative deviations in electricity price differentials. This assumption deviates from real-world scenarios, where significant heterogeneity exists in price-response characteristics across different user types.

Optimistic scenario (latent proactive response): The majority of users with elastic demand are susceptible to variations in power tariff signals. When the price difference is zero, their response intensity approaches a neutral baseline, reflecting a latent readiness to adjust consumption in response to price signals.

Table 4
Sensitivity analysis of DRM-TD model parameters.

Parameter	Inertial region	Elastic response region	Response-limited region	Regulatory effect on peak-valley difference
ν	/	$\surd$	$\diamond$	$-$
δ	/	$\times$	/	$+$
μ	/	$\underline{\Diamond}$	/	$+$
η	$\diamond$	$\diamond$	$\diamond$	$-$
$m_{p\text{-}v,t}$	/	$\times$	/	$+$
$n_{p\text{-}v,t}$	/	$\times$	/	$+$

Symbol definitions:
Impact on response regions:
/: No impact; $\diamond$: Linear impact; $\underline{\Diamond}$: Nonlinear impact.
Impact on user responsiveness estimation (Elastic response region only):
$\surd$: Parameter increase overestimates responsiveness; $\times$: Parameter increase underestimates responsiveness.
Impact on peak-valley difference:
$+$: Larger parameter → larger peak-valley difference; $-$: Larger parameter → smaller peak-valley difference.

Pessimistic scenario (response sluggishness): In comparison, Users with inelastic demand are insensitive to variations in power tariff signals and tend to maintain their inherent electricity consumption patterns. Even exposed to minor price deviations, such users remain unwilling to modify their energy usage habits. Accordingly, their baseline response falls below the neutral threshold, which reflects a pessimistic trend characterized by sluggish response behaviors. Consequently, when the price difference is zero, their baseline response lies below the neutral level, manifesting as a pessimistic "response sluggishness" tendency.

Thus, a non-zero η is necessary to capture the asymmetric response behavior observed in actual user populations.

5) Sensitivity analysis of boundary parameters.

The sensitivity of the response function to the left boundary $m_{p-v,t}$ (i.e., the threshold for entering the elastic response region) and the right boundary $n_{p-v,t}$ (i.e., the threshold triggering the response-limited region) is analyzed by considering three distinct operating regimes.

Inertial region ($0 \le \Delta\omega_{p-v,t} \le m_{p-v,t}$): in this regime, user DR behavior remains in the inertial region, where the response function is independent of the boundary parameters. Consequently, the partial derivatives with respect to both boundaries are zero:

$$\frac{\partial\widehat{\psi}_{p-v,t}}{\partial m_{p-v,t}} = \frac{\partial\widehat{\psi}_{p-v,t}}{\partial n_{p-v,t}} = 0 \tag{64}$$

Response-limited region ($n_{p-v,t} \le \Delta\omega_{p-v,t}$): in this regime, user response potential is fully exhausted, and the response function also exhibits no sensitivity to variations in the boundary parameters. Thus,

$$\frac{\partial\widehat{\psi}_{p-v,t}}{\partial m_{p-v,t}} = \frac{\partial\widehat{\psi}_{p-v,t}}{\partial n_{p-v,t}} = 0 \tag{65}$$

Elastic response region ($m_{p-v,t} \le \Delta\omega_{p-v,t} \le n_{p-v,t}$): in this regime, the response function depends explicitly on both boundaries. The partial derivative with respect to$m_{p-v,t}$is given by:

$$\frac{\partial\widehat{\psi}_{p-v,t}}{\partial m_{p-v,t}} = \frac{\psi_{max,p-v,t} - \psi_{min,p-v,t}}{2} \cdot \frac{\Delta\omega_{p-v,t} - n_{p-v,t}}{\left(n_{p-v,t} - m_{p-v,t}\right)^2} \tag{66}$$

Similarly, the partial derivative with respect to$n_{p-v,t}$can be derived as:

$$\frac{\partial\widehat{\psi}_{p-v,t}}{\partial n_{p-v,t}} = \frac{\psi_{max,p-v,t} - \psi_{min,p-v,t}}{2} \cdot \frac{m_{p-v,t} - \Delta\omega_{p-v,t}}{\left(n_{p-v,t} - m_{p-v,t}\right)^2} \tag{67}$$

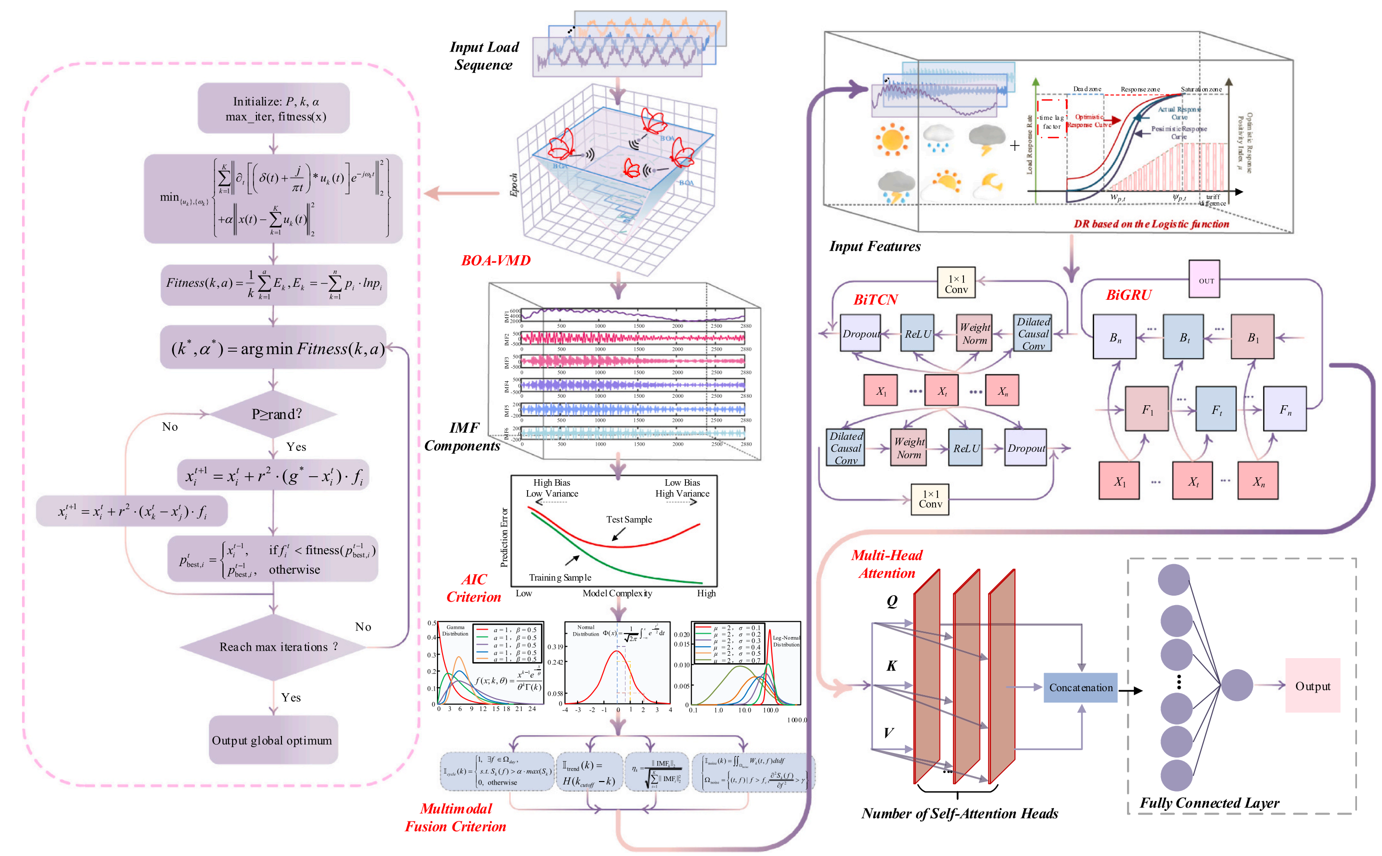


Fig. 8. Flowchart of the proposed method.

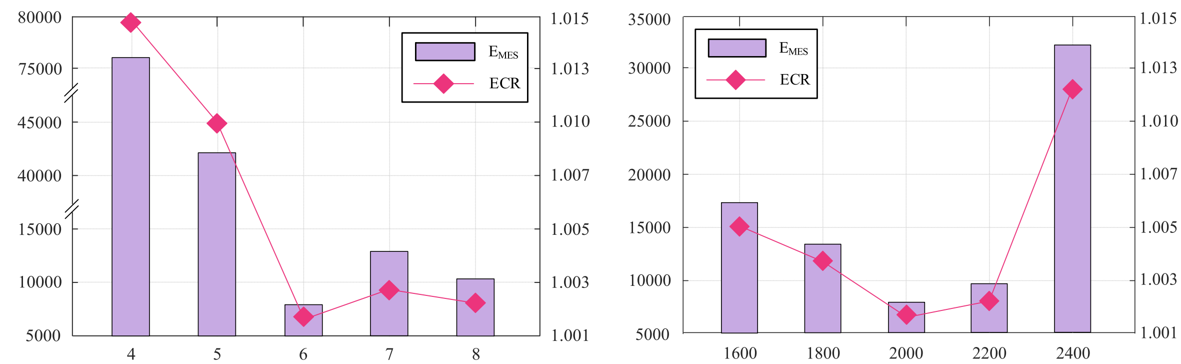
These formulations reveal that within the elastic response region, the sensitivity of the response function toward boundary parameters remains non-zero and varies linearly with the deviation between the current price gap and the opposite threshold.

The sign analysis of the partial derivatives rigorously proves that increasing either left boundary $m_{p-v,t}$ or right boundary $n_{p-v,t}$ leads to underestimation of user responsiveness and consequently a larger peak-valley difference in the load forecast, whereas decreasing these boundaries leads to overestimation of user responsiveness and a smaller peak-valley difference. This provides a theoretical basis to guide the proper calibration of model parameters: if the observed peak-valley difference is too large, $m_{p-v,t}$ or $n_{p-v,t}$ should be appropriately decreased to enhance the estimated response and reduce the peak-valley gap; conversely, if the peak-valley difference is too small, these boundaries should be increased to avoid overestimation. A summary detailing the parametric sensitivity analysis and their corresponding influences is presented in Table 4.

(6) Parameter calibration, generalizability, and limitations of the DRM-TD model.

The value of the proposed DRM-TD model lies not in obtaining a fixed set of optimal parameters, but in revealing the fundamental relationships between user behavior and model parameters, as well as in its parametric structure that can adapt to behavioral uncertainty. The generalized applicability of this framework is discussed from three complementary dimensions.

1) Calibration using real-world data: In this paper, all DRM-TD parameters are calibrated at the aggregated load level using historical load and price data from the Quanzhou grid. Specifically, the logistic function parameters are estimated via nonlinear least squares fitting, minimizing the error between the predicted load shifting quantity and the observed peak-valley load differences. The threshold$m_{p-v,t}$of the inertial region and the threshold$n_{p-v,t}$of the response-limited region are determined according to the empirical distribution of price gaps that trigger measurable load variations. The optimistic response index κ and delay factor ρ are calibrated using a grid search that maximizes the correlation between the modeled response and the actual load adjustments. This calibration process is reproducible and can be similarly applied to other regions using their own historical data.
2) Stability across different user populations: The sensitivity analysis presented in Section 3.3 demonstrates that the model's output is robust to moderate variations in each parameter. The model exhibits distinct response characteristics across different regions, which can be represented by adjusting the parameter set {ν, δ, μ, η, κ, ρ,$m_{p-v,t}$,$n_{p-v,t}$}within their respective plausible ranges. For instance, user groups with high flexibility correspond to a larger ν and a smaller δ, whereas communities with strong behavioral inertia feature an elevated ρ and a higher inertial region threshold. Thus, rather than assuming a single universal parameter set, the model provides a parametric family that can be instantiated for different user segments.
3) Advantages over alternative behavioral models: As demonstrated in the previous subsection, the logistic function outperforms linear, exponential, step, Gompertz, and arctangent formulations in characterizing the three-stage response pattern covering the inertial region, elastic response region, and response-limited region, while also ensuring symmetric responsiveness and favorable mathematical tractability. Moreover, unlike pure data-driven black-box models, the DRM-TD parameters have explicit behavioral interpretations, enabling system operators to adjust them according to observed or expected changes in user behavior.



(a) Effect of decomposition layers k on decomposition performance

(b) Effect of penalty factor r on decomposition performance

Fig. 9. Error analysis under different (k, r) combinations.

4) Limitations: Despite its advantages, the DRM-TD model has several limitations. First, the current calibration relies on aggregated load data rather than individual user-level measurements, which may mask heterogeneity within the user population. Second, the model assumes that user responses to price signals are primarily determined by price differences, while other factors such as weather, day type, and social influences are treated as external inputs. Third, the model parameters are currently calibrated using historical data and do not automatically adapt to non-stationary behavioral changes over time. Future work could incorporate online learning or Bayesian updating to track evolving user behavior.

In summary, the DRM-TD model offers a theoretically sound, behaviorally interpretable, and practically calibratable framework for DR modeling, while its limitations indicate clear directions for further improvement. When integrated with the BOA-VMD decomposition, AMFFS denoising, and BiTCN-BiGRU-MHA forecasting structure, the overall proposed short-term load forecasting framework for VPP is illustrated in Fig. 8.

4. Case study analysis

4.1. Experimental analysis of the BOA-VMD

To verify the effectiveness and accuracy of the proposed BOA-VMD algorithm, this paper uses electricity load data from Quanzhou, Fujian Province, as the dataset for experimentation. The BOA is used to optimize two key parameters of the VMD: decomposition layers k and penalty factor r. According to literature [28], setting the decomposition layers k too high can lead to overfitting due to an increased mode dimension, which also amplifies the model's sensitivity to noise in the training data. Therefore, the maximum value of k is set to 10. The optimization results show that the BOA-VMD algorithm converges to the optimal parameter combination $(k, r) = (6, 2000)$. To verify whether the obtained parameter combination is optimal, this paper uses reconstruction error and energy conservation ratio (ECR) as evaluation metrics. The reconstruction error is quantified by the difference between the original signal x and the reconstructed signal $\widetilde{x}$, as shown in Eq. (68):

$$E_{MSE}(x, \widetilde{x}) = \frac{1}{N}\sum_{i=1}^{N}(x_i - \widetilde{x}_i)^2 \tag{68}$$

where N denotes the number of data samples. The ECR evaluates the energy preservation characteristics during the decomposition and reconstruction process via Eq. (69):

$$\text{ECR} = \frac{\|\widehat{\mathbf{x}}\|^2}{\|\mathbf{x}\|^2} \tag{69}$$

where $\|\widehat{\mathbf{x}}\|^2$ and $\|\mathbf{x}\|^2$ are the energies of the reconstructed and original signals, respectively. A higher ECR value, closer to 1, indicates smaller energy loss.

Fig. 9 compares the decomposition results under different (k, r) combinations. Fig. 9(a) shows the trend of reconstruction error and energy conservation ratio as k varies with $r = 2000$ fixed, while Fig. 9(b) shows the trend as r varies with $k = 6$ fixed.

The experimental results show that the parameter combination significantly influences reconstruction performance. From Fig. 9(a), it can be observed that when the penalty factor $r = 2000$ is fixed, an increase in the decomposition layers k initially reduces the reconstruction error but then causes it to rise again. When $k = 4$, insufficient decomposition causes mode aliasing between daily and weekly periodic features, resulting in a significant increase in reconstruction error, which

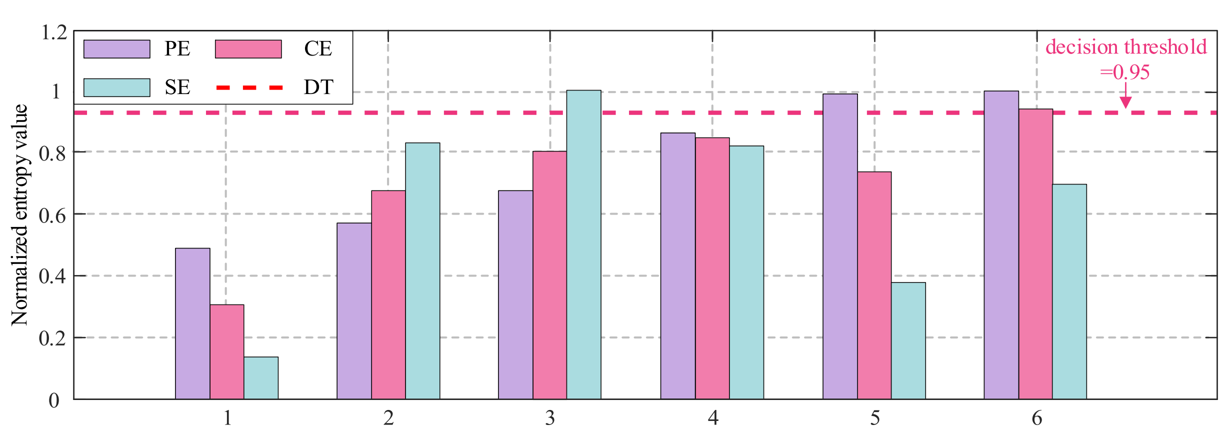


Fig. 10. Complexity evaluation results of IMF components.

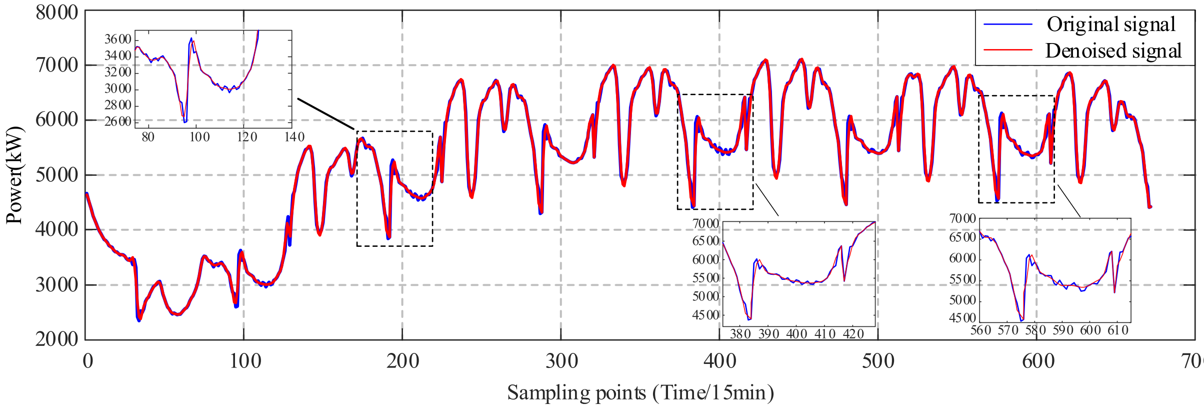


Fig. 11. Comparison of load curves before and after denoising.

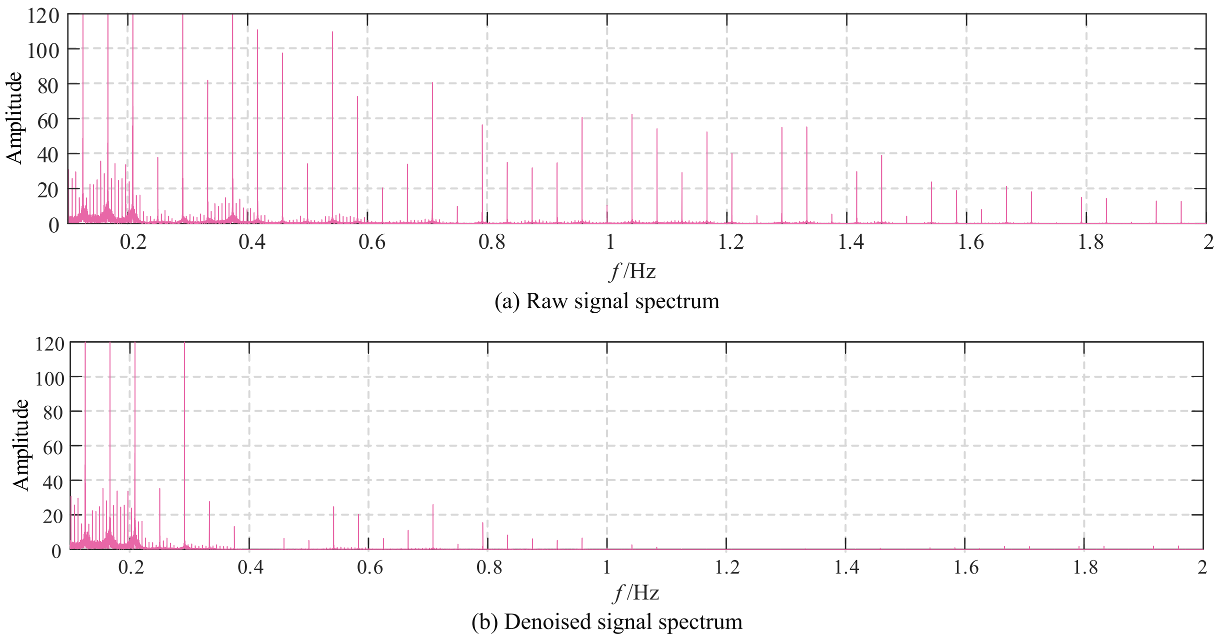


Fig. 12. Comparison of load spectrum before and after denoising.

reaches 75,742.33. At this point, the components are not fully decomposed, and the energy is concentrated in a few components, resulting in a high energy retention rate. When $k = 6$, the decomposition layers is optimal, effectively separating the trend, daily periodic features, and noise, and the reconstruction error drops to the minimum value of 9138.02. When k further increases to 7 or more, over-decomposition introduces redundant high-frequency noise components, and the optimization process diverges due to the strong constraint of $r = 2000$, leading to a rebound in the error. Moreover, by comparing the decompositions for $k = 7$ and $k = 8$, it is observed that when $k = 7$, the decomposition layers may exceed the actual mode number of the signal, leading to over-decomposition. This results in some noise or high-frequency fluctuations being mistakenly considered as valid components of the signal, thereby increasing the reconstruction error and affecting the energy conservation ratio. When $k = 8$, although more modes are decomposed, it does not reach the over-decomposition level, and both reconstruction error and energy conservation ratio improve.

Fig. 9(b) further demonstrates that when $k = 6$ is fixed, an increase in r significantly alters the reconstruction performance. When $r = 1600$, insufficient bandwidth constraints cause adjacent mode spectrums to overlap, dispersing the energy and resulting in a large reconstruction error of 17,852.52. When $r = 2000$, the constraint strength is moderate, and the mode components are clearly separated with energy concentrated in the valid components, resulting in a reconstruction error of 9138.02. However, when r is too large (e.g., $r = 2400$), the strong constraints imposed by VMD may lead to an excessively narrow modal bandwidth, failing to effectively separate noise from the true signal components. As a result, noise is incorporated into invalid or meaningless modes, causing the error to surge sharply to 33,073.39.

In conclusion, the optimized parameter combination $(k, r) = (6, 2000)$ for the BOA-VMD algorithm achieves an optimal balance between the effective separation of multi-time-scale features and energy conservation. Compared to the error at $k = 4$, this parameter combination reduces the reconstruction error by 87.9%. Additionally, by balancing the constraint strength of the penalty factor r, the algorithm effectively avoids modal aliasing caused by insufficient constraints (e.g., $r = 1600$) or divergence risks caused by excessive constraints (e.g., $r = 2400$), ensuring the stable convergence of the VMD algorithm.

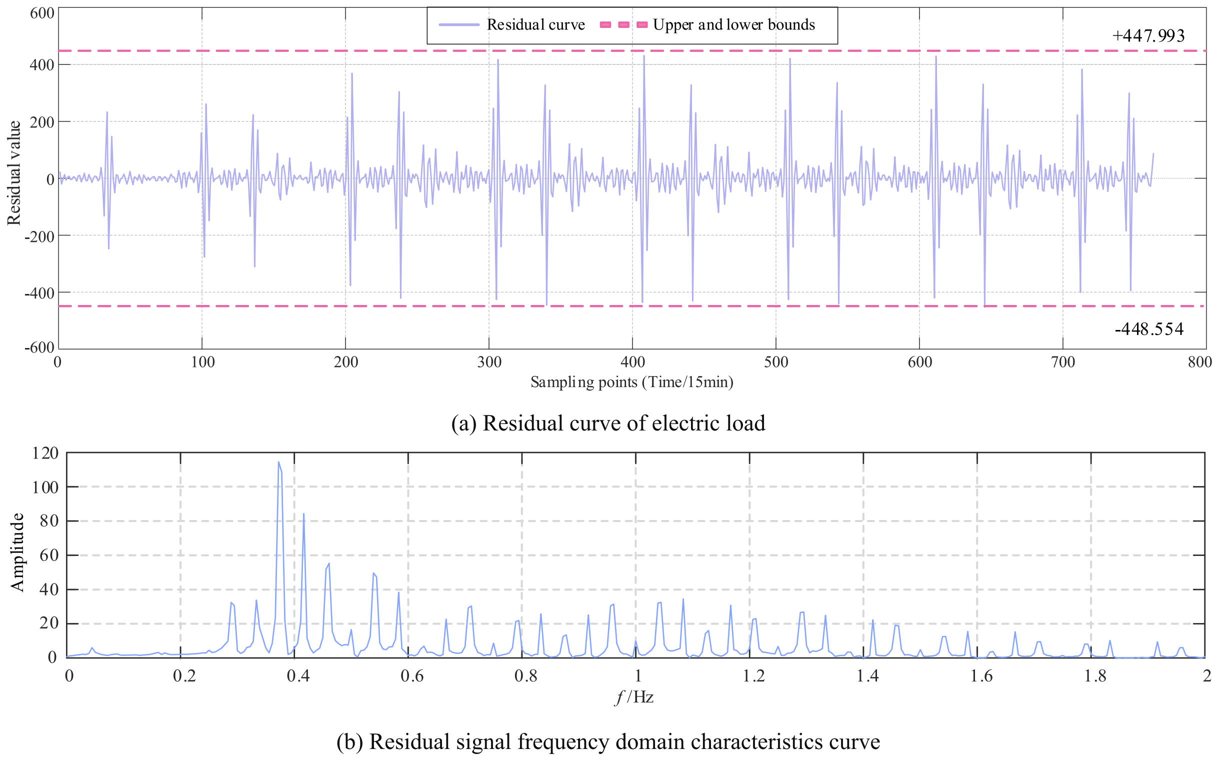


Fig. 13. Time-frequency domain evaluation of denoising effect.

4.2. Verification of the AMFFS denoising method

To verify the effectiveness of the AMFFS method proposed in Section 3.2 and its impact on improving prediction accuracy, this paper conducts experimental analysis from multiple dimensions, including time-frequency domain joint analysis, signal reconstruction quality assessment, and residual characteristics study.

Fig. 10 shows the quantitative evaluation results of the complexity of six IMF using composite entropy (CE). The results indicate that IMF1 has a CE of 0.349, significantly lower than the decision threshold of 0.95, suggesting it primarily contains low-frequency trend signals, such as daily or seasonal variations. The entropy values of IMF2 to IMF5 are relatively high, corresponding to mid-frequency periodic components, mainly reflecting weekday load fluctuations. In contrast, IMF6 has composite entropy of 0.954, significantly exceeding the threshold of 0.95, indicating that it is dominated by high-frequency noise. This entropy distribution pattern aligns with the typical characteristics of electricity load data, where low-frequency trends dominate and high-frequency noise is superimposed. Therefore, by retaining IMF1–IMF5 and removing IMF6, the signal can be reconstructed while effectively filtering out high-frequency noise, preserving the key low- and mid-frequency information.

Fig. 11 visually presents the comparison of load curves before and after denoising in the time domain. The analysis shows that the signal processed by the AMFFS method exhibits a significant advantage in suppressing short-term fluctuations. Compared to the original signal's fluctuations, the denoised curve achieves excellent smoothing characteristics while maintaining the basic load shape. The AMFFS method effectively eliminates short-term abnormal fluctuations caused by random interference while preserving the primary trend features reflecting system operation.

To comprehensively evaluate the denoising effect of the proposed method in the frequency domain, Fig. 12 compares the power spectral density (PSD) distributions of the signal before and after denoising.

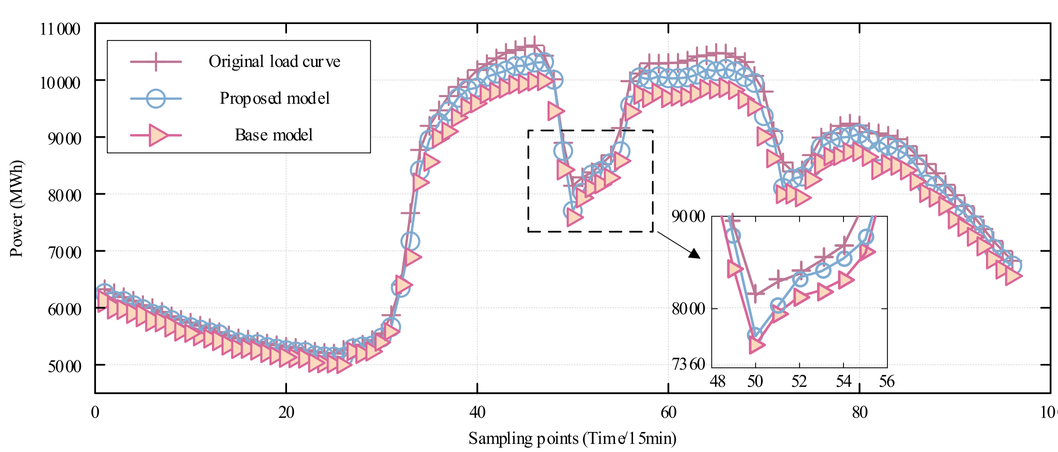


Fig. 14. Comparison of load forecasting before and after denoising.

Table 5
Comparison of forecasting performance before and after load data denoising.

Predictive model	R^2	MAE	RMSE	MAPE
Base model	0.958	339.61	382.34	4.17
Proposed model	0.972	231.89	305.85	3.31

Table 6
Time-of-use pricing parameters.

Period type	Time periods	Price/ Yuan
Peak period	10:00–12:00; 15:00–20:00; 21:00–22:00	0.842
Flat period	08:00–10:00; 12:00–15:00; 20:00–21:00; 22:00–24:00	0.533
Valley period	0:00–08:00	0.197

From the comparison of the main frequency positions, it is evident that the main frequency components of the denoised signal remain consistent with the original signal, indicating that the smoothing process does not interfere with the core periodic features of the signal. Further analysis of the high-frequency region shows a significant reduction in disturbances and energy amplitude, weakening the high-frequency noise caused by random interference. Notably, the energy distribution in the low-frequency band closely matches that of the original signal, indicating that the low-frequency trend components are effectively preserved. Compared to the original signal spectrum, the denoised spectrum is clearer, with the main frequency components more prominent and noise components significantly reduced, achieving effective noise suppression while preserving the signal's essential features. This result confirms the superior denoising performance and signal fidelity of the proposed method in the frequency domain. Fig. 13 further quantifies the denoising effectiveness by plotting the residual curves and their frequency-domain characteristics.

The analysis of the residual curve in Fig. 13(a) shows that the residual values of the denoised signal are stably distributed within the ±3σ range, indicating that the difference between the denoised signal and the original electric load signal is effectively controlled within a small range. This demonstrates that the denoising process significantly improves data stability. Further analysis of the frequency - domain characteristics in Fig. 13(b) demonstrates that the spectral energy of the residual signal is concentrated in the low-frequency region, with energy in the high-frequency part significantly reduced. This frequency domain feature indicates that the AMFFS method, through the multimodal feature denoising mechanism, effectively eliminates high-frequency noise while fully preserving the low-frequency trend and periodic characteristics of the original signal.

Table 7
Forecasting results of four models under TOU.

Time	Actual load / MW	Load forecast error / MW			
		Model 1	Model 2	Model 3	Model 4
0:00	6326.131	−355.107	−372.214	−197.186	−191.624
1:00	6046.680	−338.081	−352.616	−115.474	37.813
2:00	5790.228	−242.015	−261.057	26.608	82.111
3:00	5571.339	−372.606	−231.715	−195.542	−149.539
4:00	5402.350	−434.006	−261.363	−184.367	−184.139
5:00	5296.274	−425.180	−329.828	−149.401	44.965
6:00	5195.618	−356.978	−264.627	−126.837	149.138
7:00	5433.530	−340.231	−234.320	−148.530	−175.186
8:00	7664.381	−314.890	−194.231	−192.677	−51.767
9:00	9718.424	−280.711	−113.652	−118.580	−162.533
10:00	10,281.515	−279.354	−107.937	−133.059	105.419
11:00	10,584.934	−238.015	−70.479	−100.030	−106.813
12:00	8898.739	−235.907	−83.143	−108.869	−43.895
13:00	8523.090	−227.046	−90.140	−116.806	77.094
14:00	10,116.396	−195.566	−52.257	−119.513	−40.313
15:00	10,303.396	−273.638	−89.753	−181.739	−107.599
16:00	10,476.720	−179.603	−45.657	−90.906	−157.017
17:00	10,073.187	−309.814	−142.092	−107.793	5.115
18:00	8395.756	−386.892	−222.962	−120.002	−2.410
19:00	9067.730	−328.769	−273.417	−178.427	−185.423
20:00	9087.520	−186.181	−2.686	52.396	27.575
21:00	8839.159	−450.234	−255.139	−263.758	−223.250
22:00	8160.590	−345.012	−227.493	−135.277	−74.662
23:00	7384.234	−300.459	−149.129	−55.423	−17.621

In conclusion, the proposed AMFFS method not only significantly reduces noise interference in load data but also accurately preserves the primary feature components of the signal, providing a high-quality data foundation for subsequent prediction tasks.

4.3. *Effectiveness evaluation of the AMFFS method and performance comparison in the TOU scenario*

(1) Evaluation of load forecasting improvement based on the AMFFS method.

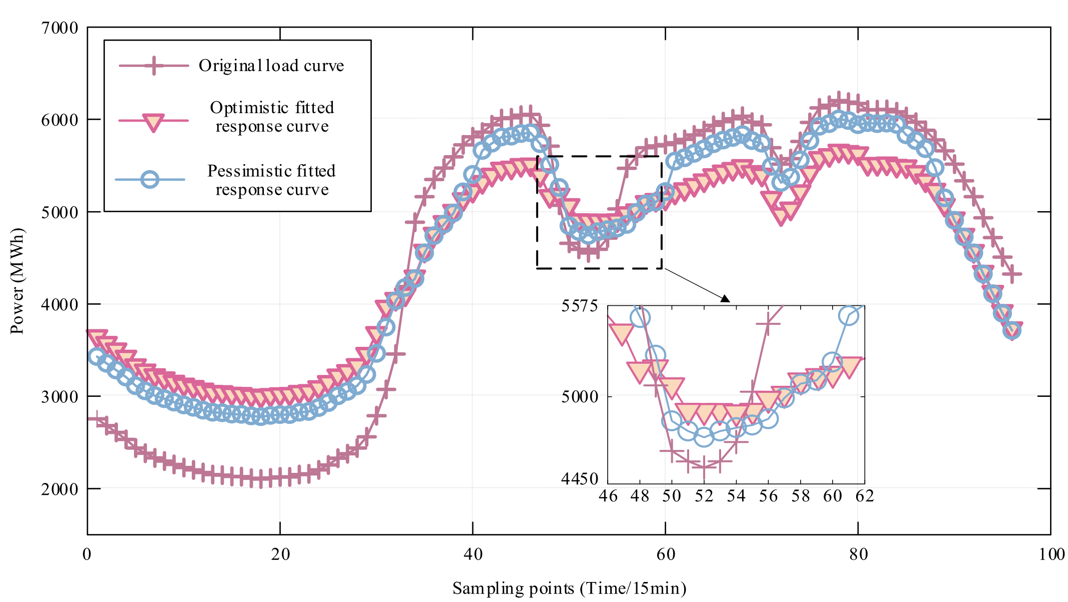


Fig. 15. Forecasting results based on DRM-TD.

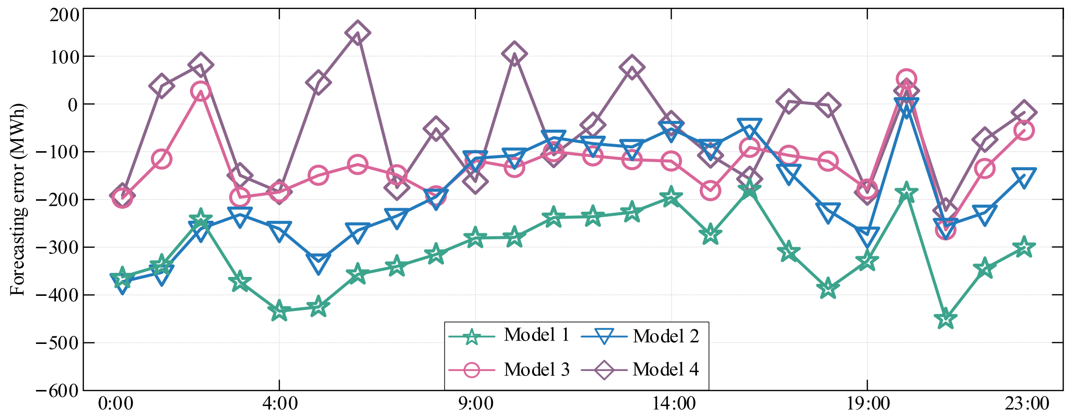


Fig. 16. Comparison of forecasting errors across four models under TOU.

To systematically assess the improvement effect of the proposed AMFFS method on electricity load forecasting, a controlled experiment was designed. Forecasting models were constructed using the original data and AMFFS-processed data, denoted as the base model and proposed model, respectively. The models were compared using performance metrics such as mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and coefficient of determination (R^2).

Fig. 14 presents a comparison of the typical daily load forecasting results for the Quanzhou grid. The research finds that the training data processed using the AMFFS method significantly improves the model's forecasting accuracy. Compared to the model trained on the original data, the model trained with AMFFS-processed data shows a substantial improvement in both load curve fitting and the ability to capture key features. Table 5 further quantifies the performance differences between the two models.

As shown in Table 5, the forecasting model optimized by the AMFFS method demonstrates multidimensional improvements in key performance indicators. In terms of goodness-of-fit, the R^2 increases from 0.958 to 0.972, indicating enhanced capability in capturing load variation characteristics. Regarding error control, the MAE decreases by 31.72%, while the RMSE and MAPE reduce by 20% and 20.62%, respectively. Experimental results confirm that the AMFFS method not only effectively suppresses random noise but also significantly improves the accuracy of load forecasting models.

(2) Performance comparison of the forecasting models in the TOU scenario.

To systematically validate the role of the DR model with time delay (DRM-TD) in load forecasting, four sets of comparative models were built using the Quanzhou grid dataset:

Model 1: The standard BiTCN-BiGRU-Multi-Head Attention model based on the features in Eq. (38).

Model 2: The BiTCN-BiGRU-Multi-Head Attention model combined with AMFFS.

Model 3: The DRM-TD framework BiTCN-BiGRU-Multi-Head Attention model based on features in Eq. (48).

Model 4: The BiTCN-BiGRU-Multi-Head Attention model combined with AMFFS and DRM-TD.

Based on the historical load data of the Quanzhou grid, a TOU incentive framework was established. The time periods and pricing parameters are shown in Table 6.

For better visualization, typical daily load data was selected for model validation and analysis. The specific steps are as follows:

Table 8
Comparative performance analysis of four models under TOU.

Predictive model	R^2	MAE	RMSE	MAPE (%)	AMFFS	DRM-TD
Mode1	0.956	335.96	392.07	4.04	×	×
Mode2	0.970	274.54	335.52	3.52	√	×
Mode3	0.974	264.21	296.26	3.29	×	√
Mode4	**0.995**	**74.16**	**99.79**	**0.94**	√	√

1. Construct a DRM-TD model based on user behavior-driven mechanisms, using the Logistic function to model the nonlinear features of user response behavior.
2. Use model 4 for load forecasting experiments based on the DRM-TD modeling results. The comparison between the predicted curve and actual load data is shown in Fig. 15.

Fig. 15 shows that the Logistic function-based response mechanism model effectively reveals the regulatory effects of TOU policies on the grid load distribution. By simulating user behavior response characteristics, this model validates the effectiveness of price signals in guiding load shifts. During the peak periods defined in Table 7 (12:00–14:00; 19:00–22:00), the price incentives encourage users to actively adjust their electricity consumption, resulting in a significant decrease in peak load demand. Conversely, during the valley periods (01:00–07:00; 23:00–24:00), the lower electricity price stimulates users to increase their electricity consumption, facilitating the effective transfer of load to the off-peak periods. Notably, under optimistic response assumptions, users exhibit higher responsiveness, with a nonlinear increase in user sensitivity to price incentives. This finding not only validates the feasibility of TOU in peak shaving and valley filling but also highlights the crucial role of user behavior in DR modeling.

To verify the superiority of the proposed model, load forecasting was conducted for a typical day in the target grid using Models 1 through 4, based on the TOU scenario defined in Table 7. To ensure clarity and intuitiveness of the data, only the corresponding de-normalized load values at the hour marks are presented. The forecasting results and error comparisons are shown in Table 7 and Fig. 16.

Analysis of Table 7 reveals that models 1 and 2 have significant limitations in load forecasting under the TOU scenario, particularly when capturing the load fluctuations induced by price incentives. For instance, during typical time periods, at 17:00, the prediction errors for models 1 and 2 are −309.814 MW and − 142.092 MW, respectively. At 22:00, the errors are −345.012 MW and − 227.493 MW, indicating that these models struggle to effectively capture the nonlinear relationship between price fluctuations and load response. In contrast, Models 3 and 4 show significant forecasting advantages. Model 3, which introduces DRM-TD, keeps the maximum absolute error within 263.76 MW,

Table 9
Sensitivity analysis of weight coefficient *a*.

a	R^2	MAE	RMSE	MAPE (%)
0.3	0.955	342.38	398.14	4.11
0.4	0.977	243.13	276.49	2.72
0.5	0.994	75.21	95.42	1.01
0.6	0.982	147.46	175.41	1.82
0.7	0.973	265.23	286.58	3.35

significantly narrowing the error fluctuation range compared to the traditional models. Further integration of the AMFFS method in Model 4 leads to a substantial improvement in forecasting accuracy. At 17:00, the error is only 5.115 MW, and at 18:00, the absolute error further decreases to 2.410 MW. This demonstrates the critical role of the synergistic optimization of DRM-TD and AMFFS methods in improving model prediction accuracy.

Fig. 16 visually presents the distribution of prediction errors for the four models under the TOU scenario. The error curves of model 1 and model 2 exhibit significant volatility, with their absolute error values exceeding 200 MW during the 18:00–19:00 period marked by abrupt load fluctuations. This highlights their limitations in capturing load growth trends and mitigating prediction errors. In contrast, the error volatility of Model 3 and Model 4 is markedly reduced, with overall error values remaining largely within ±200 MW, demonstrating their greater precision in forecasting load growth trends.

Table 8 further quantifies the performance differences among the four models using evaluation metrics. The results show that traditional deep learning models exhibit significant limitations in the TOU scenario. Model 1 has an MAE of 335.96 MW and an MAPE of 4.04%, while Model 2 has an MAE of 274.54 MW and an MAPE of 3.52%. Both models do not exceed the performance threshold of $R^2 = 0.97$. These prediction errors significantly surpass reasonable thresholds, especially during the period of load fluctuations (17:00–18:00), which reveals an obvious under fitting phenomenon. This validates the inadequate ability of these models to capture load features in the price incentive scenario. In contrast, Models 3 and 4 show significant advantages. Model 3, by introducing DRM-TD, reduce MAE to 264.21 MW, a 21.36% decrease compared to Model 1, while the MAPE drops to 3.29%, and R^2 exceeds the 0.97 threshold. Model 4, which further integrates the AMFFS method, achieves an MAE of 74.16 MW, and R^2 reaches 0.995. The results suggest that under the TOU scenario, the hybrid model combining AMFFS and DRM-TD outperforms traditional deep learning models in capturing load dynamics and enhancing forecasting accuracy.

4.4. Sensitivity analysis of weight coefficient (a) in composite entropy

The composite entropy S_c defined in Eq. (10) introduces a weight coefficient a to balance the contribution of permutation entropy (PE) and sample entropy (SE). To investigate the impact of a on the final forecasting performance and to justify the choice of $a = 0.5$, a sensitivity analysis was conducted by varying aa in the range of 0.3–0.7 at an interval of 0.1, whereas all remaining model parameters were maintained unchanged. The experiments were performed under the TOU scenario using the same dataset and evaluation metrics as in Section 4.3.

Table 9 summarizes the forecasting performance for different a value. It is evident that the model achieves the best performance when $a = 0.5$, with $R^2 = 0.994$, MAE = 75.21 MW, RMSE = 95.42 MW, and MAPE = 1.01%. Both underestimating the weight of PE, for example $a = 0.3$, and overestimating it, for example $a = 0.7$, lead to a significant deterioration in prediction accuracy. Specifically, when $a = 0.3$, which favors SE, the MAE increases to 342.38 MW and the MAPE rises to 4.11%, indicating that insufficient sensitivity to high-frequency noise causes the model to retain random fluctuations, thereby degrading its ability to capture underlying load patterns. Conversely, when $a = 0.7$, which favors PE, the MAE becomes 265.23 MW and the MAPE reaches 3.35%, suggesting that excessive emphasis on permutation entropy may mistakenly identify some meaningful high-frequency load variations as noise, leading to over-smoothing and loss of critical features.

These results demonstrate that the two entropy measures play complementary roles in characterizing signal complexity: PE excels at detecting short-term randomness and noise, while SE preserves long-term trend information. A balanced weighting $a = 0.5$ achieves an optimal trade-off, enabling the AMFFS method to suppress random noise without distorting essential load characteristics. Moreover, the performance degradation is symmetric around $a = 0.5$, confirming that the proposed composite entropy is robust to moderate deviations. Therefore, fixing $a = 0.5$ is not an arbitrary choice but a well-justified default that ensures consistent and accurate forecasting across different datasets and scenarios.

Table 10
Dynamic risk period division and pricing strategy.

Risk level	Time period	Price adjustment margin	Core objective
High-risk	10:00–12:00, 15:00–17:00, 20:00–22:00	+58%	Suppress peak demand
Medium-risk	08:00–09:00, 13:00–14:00, 18:00–19:00, 23:00–24:00	0%	Maintain market stability
Low-risk	01:00–07:00	−63%	Incentivize load shifting

4.5. Comparison of model forecasting performance under RTP

The formulation of RTP requires a comprehensive consideration of multi-dimensional factors such as users' historical load data trends, load dynamic characteristics, and power market supply-demand fluctuations, in order to establish a scientifically sound and reasonable pricing scheme. This paper employs the following steps to design the real-time electricity pricing model:

(1) Data preprocessing and feature extraction.

The raw 15-min interval load data is first aggregated into hourly average load values to reduce the dimensionality of the data. Next, the cyclic differencing method is used to extract the load rate of change between adjacent periods, thereby quantifying the load fluctuation characteristics. On this basis, a dual normalization processing mechanism is further constructed, mapping the load level and rate of change to a normalized interval [0, 1]. The calculation formula is as follows:

$$\begin{cases} L_h = \dfrac{L - L_{min}}{L_{max} - L_{min}} \\ \Delta L_h = \dfrac{\Delta L - \Delta L_{min}}{\Delta L_{max} - \Delta L_{min}} \end{cases} \tag{70}$$

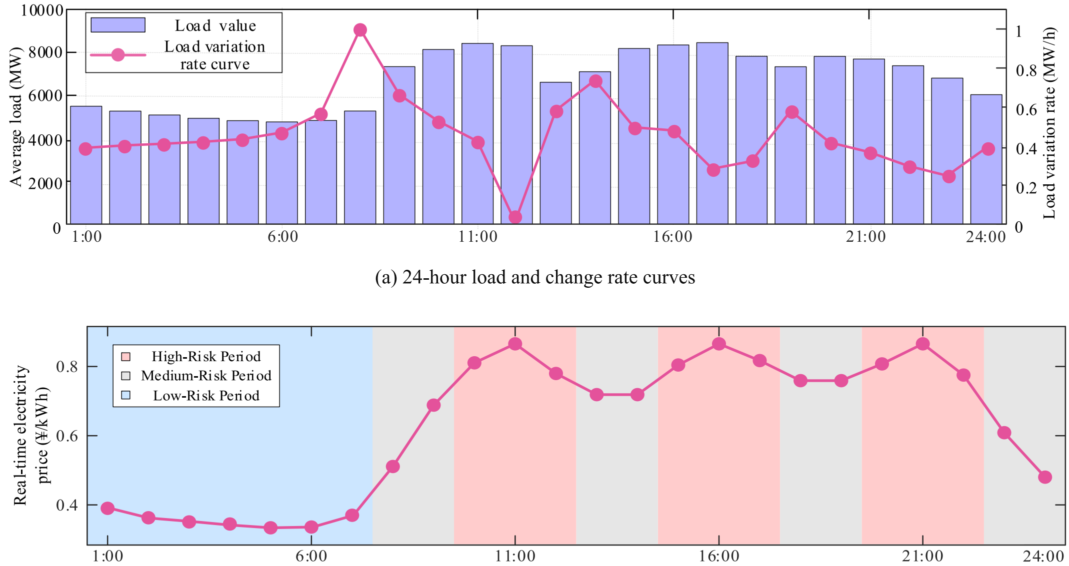
where L_h is the normalized load value, L is the original load value, L_{max} and L_{min} are the maximum and minimum load values, respectively. Similarly, ΔL_h is the normalized load change rate, ΔL is the original load change rate, and ΔL_{max} and ΔL_{min} are the maximum and minimum load change rates, respectively.

(2) Dual-factor dynamic model construction.

Based on the standardized load value and change rate features, this paper constructs a load-fluctuation dual-factor pricing model. The calculation formula is as follows:

$$P_h^{base} = P_{min} + (P_{max} - P_{min})\cdot[\varphi\cdot L_h + (1-\varphi)\cdot\Delta L_h] \tag{71}$$

where$\varphi$is the weight coefficient, set asφ= 0.5 in this paper to balance the sensitivity of price signals to load levels and the ability to capture short-



(a) 24-hour load and change rate curves

(b) Risk period division and RTP

Fig. 17. Analysis of power load Characteristics and dynamic division of RTP.

Table 11
Forecasting results of four models under RTP.

Time	Actual load / MW	Load forecast error / MW			
		Model 1	Model 2	Model 3	Model 4
0:00	6326.131	−421.969	−429.288	−440.577	−108.532
1:00	6046.680	−358.033	−161.333	−423.066	−69.949
2:00	5790.228	−299.514	−357.196	−176.626	−8.852
3:00	5571.339	−398.764	−357.951	−353.111	−69.006
4:00	5402.350	−425.181	−368.434	−317.916	−87.002
5:00	5296.274	−423.446	−385.019	−359.122	−114.050
6:00	5195.618	−351.112	−309.581	−296.586	−32.173
7:00	5433.530	−266.518	−296.117	−287.009	−74.407
8:00	7664.381	−274.037	−260.701	−262.878	−67.539
9:00	9718.424	−204.076	−204.883	−217.057	−53.970
10:00	10,281.515	−188.160	−232.661	−215.229	−73.006
11:00	10,584.934	−153.223	−218.945	−174.094	−48.934
12:00	8898.739	−166.280	−233.603	−191.175	−56.004
13:00	8523.090	−160.365	−226.277	−183.714	−51.415
14:00	10,116.396	−145.568	−202.309	−152.815	−37.616
15:00	10,303.396	−216.245	−302.026	−240.937	−121.617
16:00	10,476.720	−224.956	−110.680	−187.916	−15.489
17:00	10,073.187	−254.541	−241.063	−204.578	−32.306
18:00	8395.756	−346.011	−285.158	−296.264	−65.468
19:00	9067.730	−356.251	−280.517	−351.598	−67.814
20:00	9087.520	−87.555	−283.309	−108.172	79.633
21:00	8839.159	−342.477	−428.057	−314.181	−279.228
22:00	8160.590	−320.065	−295.085	−270.753	−64.747
23:00	7384.234	−277.473	−230.393	−204.604	2.441

term fluctuation risks. This model uses a dynamic weight allocation mechanism, which maintains the effectiveness of the electricity price response to load while introducing the load rate of change to quantify the risk of sudden load changes, enabling refined modeling of DR behavior.

(3) Dynamic time period division and price function design.

Based on the typical daily load curve characteristics of Fujian province and the requirements of RTP policies, this paper proposes a real-time pricing mechanism that integrates load risk levels and dual-factor pricing. First, based on historical load data and user behavior patterns, the 24-h period is divided into three risk levels, and differentiated price adjustment strategies are set accordingly. The specific time periods and price strategies are shown in Table 10.

Based on the above analysis, the RTP conversion function is constructed by mapping the standardized load level and rate of change to the risk periods. The conversion formula is as follows:

$$P_{real-time}(t) = \begin{cases} 1.58 \times P_h^{base} & t \in T_{h-r} \\ P_h^{base} & t \in T_{m-r} \\ 0.37 \times P_h^{base} & t \in T_{l-r} \end{cases} \tag{72}$$

where $T_{h\text{-}r}$, $T_{m\text{-}r}$, $T_{l\text{-}r}$ represent the high, medium, and low-risk time periods, respectively.

Based on the above analysis, the temporal characteristics of power load and the dynamic classification of RTP are illustrated in Fig. 17.

From Fig. 17(a), we can see that the load peaks are concentrated in the periods of 10:00–12:00, 15:00–17:00, and 20:00–22:00, while the load valleys occur between 01:00 and 07:00. Notably, the load change rate reaches its peak during the 07:00–08:00 period, indicating significant supply-demand contradictions during this time. Fig. 17(b) further shows the risk period division and corresponding RTP based on load levels, with red regions representing high-risk periods, gray regions representing medium-risk periods, and blue regions representing low-risk periods. Through differentiated pricing, user behavior can be effectively guided, promoting the shift of peak loads to off-peak periods and optimizing the allocation of electricity resources.

Based on the above pricing regulation framework, the proposed model's forecasting performance under dynamic pricing scenarios is evaluated by comparing Models 1 through 4 against the typical daily load of the target grid. To ensure clarity and intuitiveness of the data, only the corresponding de-normalized load values at the hour marks are presented. The forecasting results and error distribution are shown in Table 11 and Fig. 18.

Table 11 shows that the forecast errors of Models 1 and 2 significantly exceed ±200 MW across multiple time periods. For instance, at 00:00, the errors are −421.969 MW and − 429.288 MW, and at 02:00, the errors are −299.514 MW and − 357.196 MW, indicating that these models fail to adequately capture load fluctuations driven by electricity prices. In contrast, Model 3, which incorporates DRM-TD, reduces the error to −176.626 MW at 02:00, marking a 50.55% improvement over Model 2. Further, after introducing the AMFFS method in Model 4, the error is further reduced to −8.852 MW, demonstrating the superiority of

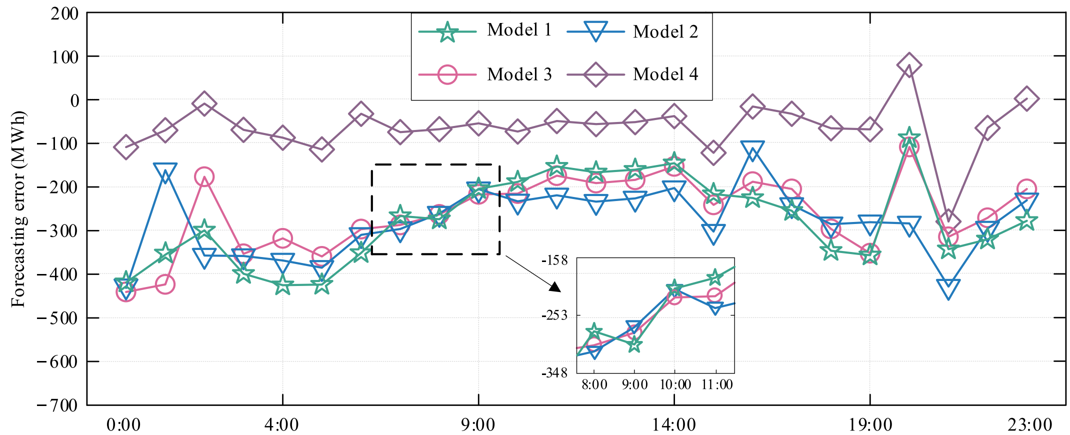


Fig. 18. Comparison of prediction errors of four models under RTP.

Table 12
Comparative performance analysis of four models under RTP.

Predictive model	R^2	MAE	RMSE	MAPE (%)	AMFFS	DRM-TD
Mode1	0.959	340.15	371.41	4.25	×	×
Mode2	0.971	253.55	314.79	3.41	√	×
Mode3	0.983	138.49	167.24	1.79	×	√
Mode4	0.994	75.21	95.42	1.01	√	√

the AMFFS method. Notably, during high-risk periods at 11:00 and 22:00, Model 4 reduces errors to −48.934 MW and − 64.747 MW, respectively, representing a 71.89% and 76.09% reduction compared to Model 3. This significant optimization suggests that the introduction of DRM-TD not only improves the model's ability to adapt to regular load fluctuations but also enhances prediction accuracy in high-risk scenarios driven by RTP, highlighting its core role in modeling user behavior under dynamic pricing mechanisms.

Fig. 18 visually presents the prediction error distribution of the four models under the RTP scenario. The results show that the error fluctuation ranges of Models 1 and 2 are significantly greater than those of Models 3 and 4, indicating inadequate modeling capability for load fluctuations. Notably, the error curves of Model 4 consistently remain lower than those of other models, and its predictions align most closely with actual load curves, confirming that the integration of DRM-TD and AMFFS methods significantly improves model prediction accuracy.

Table 12 quantifies the performance differences among the four models under a RTP mechanism. The results show that the MAE for Models 1 and 2 are 340.15 MW and 253.55 MW, with MAPE values of 4.25% and 3.41%, respectively, indicating relatively high forecast errors. In contrast, Model 3, incorporating DRM-TD, reduces the MAE to 138.49 MW, a 59.29% improvement over Model 1, and increases R^2 to 0.983. Model 4, further enhanced with the AMFFS method, reduces the MAE to 75.21 MW, a 45.69% improvement over Model 3, with R^2 reaching 0.994. These significant improvements in the quantifiable metrics highlight the synergistic effects of DRM-TD and AMFFS: DRM-TD enhances the dynamic modeling of user response behaviors, while AMFFS suppresses random noise in the load data, improving model robustness. The coupling of these two methods within the model architecture leads to a synergistic improvement in both prediction accuracy and robustness.

To further quantify the optimization effects of DRM-TD and AMFFS methods under different pricing mechanisms, Table 12 provides a systematic comparison of model performance under TOU and RTP scenarios. The gain difference (G_D) reflects the relative change in performance improvements between the two pricing mechanisms. It quantifies the sensitivity of the method to dynamic changes in the pricing mechanism. A $G_D > 0$ indicates a higher performance improvement under RTP than under TOU while a $G_D < 0$ indicates the opposite. The calculation formula for G_D is given in Eq. (73):

$$G_D = \frac{M_{RTP} - M_{TOU}}{M_{TOU}} \times 100\% \tag{73}$$

where M_{RTP} and M_{TOU} represent the relative errors under RTP and TOU, respectively.

Table 13
Comparative analysis of model optimization effects under multiple electricity price mechanisms.

Optimization dimension	Metric	TOU scenario (M_{TOU})	RTP scenario (M_{RTP})	Gain difference (G_D)	Adaptation coefficient (A_c)
Baseline model *(Mode1)*	R^2	0.956	0.959	+0.31	1.003
	MAE (MW)	335.96	340.15	+1.25	1.012
	RMSE (MW)	392.07	371.41	−5.27	0.947
	MAPE (%)	4.04	4.25	+5.20	1.052
AMFFS optimization *(Mode2 vs Mode1)*	$\Delta R^2\uparrow$ (%)	+1.46	+1.25	−14.38	0.856
	ΔMAE↓ (%)	−18.28	−25.46	+39.28	1.393
	ΔRMSE↓ (%)	−14.42	−15.24	+5.69	1.057
	ΔMAPE↓ (%)	−12.87	−19.76	+53.54	1.535
DRM-TD optimization *(Mode3 vs Mode1)*	$\Delta R^2\uparrow$ (%)	+1.88	+2.50	+32.98	1.330
	ΔMAE↓ (%)	−21.36	−59.29	+177.57	2.776
	ΔRMSE↓ (%)	−24.44	−54.97	+124.92	2.249
	ΔMAPE↓ (%)	−18.56	−57.88	+211.85	3.119
Integrated optimization *(Mode4 vs Mode1)*	$\Delta R^2\uparrow$ (%)	+4.08	+3.65	−10.54	0.895
	ΔMAE↓ (%)	−77.93	−77.89	−0.05	0.999
	ΔRMSE↓ (%)	−74.55	−74.31	−0.32	0.997
	ΔMAPE↓ (%)	−76.73	−76.24	−0.64	0.994

Table 14
Sensitivity analysis of weight coefficient φ.

φ	R^2	MAE	RMSE	MAPE (%)
0.3	0.982	147.81	175.46	1.82
0.4	0.989	106.12	133.02	1.37
0.5	0.995	74.16	99.79	0.94
0.6	0.985	127.30	155.11	1.65
0.7	0.976	233.41	264.06	2.88

The adaptability coefficient (A_c) measures the relative advantage of the proposed method under different pricing mechanisms. An $A_c > 1$ indicates that the solution is more advantageous under RTP, while an $A_c < 1$ indicates a greater advantage under TOU. The formula for A_c is given in Eq. (74):

$$A_c = \frac{M_{RTP}}{M_{TOU}} \tag{74}$$

Table 13 provides a multidimensional quantitative analysis, revealing that DRM-TD delivers significantly greater optimization gains in RTP scenarios compared to the independent contribution of AMFFS. Specifically, under RTP, DRM-TD reduces the MAE by 59.29%, whereas the reduction under TOU is only 21.36%. Furthermore, its corresponding A_c reaches 2.776, confirming the nonlinear response capability of DRM-TD to dynamic price signals. In contrast, although the AMFFS method shows improvements in both pricing scenarios, the gain magnitude varies: under TOU, the MAE and RMSE decrease by 18.28% and 14.42%, respectively, while under RTP, the MAE and RMSE decrease by 25.46% and 15.24%. The G_D for AMFFS in the RTP scenario reaches 39.28%, with an A_c of 1.393, both significantly higher than in the TOU scenario. This indicates that AMFFS is better suited for the volatility suppression requirements in RTP scenarios. It is important to note that while AMFFS can reduce the MAE by 25.46% in RTP scenario, its G_D is only 22.12% of DRM-TD's G_D, and its A_c of 1.393 is much lower than DRM-TD's 2.776. The core reason for this discrepancy is that DRM-TD, by dynamically modeling user response behavior, can accurately capture the nonlinear adjustments in load caused by RTP fluctuations. In contrast, AMFFS, as a volatility suppression technique, can only reduce errors by filtering out random noise and cannot effectively model user behavior responses have driven by price signals.

In conclusion, the introduction of DRM-TD is a key factor in improving model prediction accuracy in RTP scenarios. Model 4, by integrating the AMFFS technique and DRM-TD behavior modeling mechanism, significantly reduces error metrics, demonstrating the prediction reliability and robustness of the proposed model in complex pricing scenarios. This approach not only provides decision support for the fine-tuned regulation of virtual power grids but also highlights the necessity of collaborative optimization between behavior modeling and data noise reduction at the mechanism level.

(4) Sensitivity analysis of weight coefficient φ in dual-factor dynamic pricing model.

The dual-factor dynamic pricing model defined in Eq. (71) introduces a weight coefficient φ to balance the contributions of normalized load level L_h and normalized load change rate ΔL_h. To investigate the impact of φ on forecasting performance and to justify the selection of $\varphi = 0.5$, we conducted a sensitivity analysis by varying φ from 0.3 to 0.7 with a step of 0.1, while keeping all other model parameters fixed. The experiments were performed under the RTP scenario using the proposed model that integrates both AMFFS and DRM-TD.

Table 14 summarizes the forecasting performance for different φ values. The model achieves the best performance when $\varphi = 0.5$, with $R^2 = 0.995$, MAE = 74.16 MW, RMSE = 99.79 MW, and MAPE = 0.94%. Both lower and higher φ values lead to noticeable accuracy degradation. Specifically, when $\varphi = 0.3$, which places less emphasis on load level and more on change rate, the MAE increases to 147.81 MW and the MAPE rises to 1.82%. This indicates that over-sensitivity to short-term load fluctuations cause the price signal to react excessively to transient changes, thereby distorting the demand response modeling. Conversely, when $\varphi = 0.7$, which favors load level over change rate, the MAE becomes 233.41 MW and the MAPE reaches 2.88%. In this case, the pricing mechanism becomes sluggish in responding to sudden load variations, leading to inadequate peak-shaving and valley-filling effects, and ultimately reducing forecasting accuracy.

These results demonstrate that the load level and its change rate play complementary roles in real-time pricing design. The load level captures the steady-state demand, while the change rate reflects short-term risk and volatility. A balanced weighting $\varphi = 0.5$ achieves an optimal trade-off, enabling the DRM-TD model to accurately capture both routine load patterns and abrupt behavioral changes induced by price incentives. Moreover, the performance degradation is roughly symmetric around $\varphi = 0.5$, confirming that the proposed dual-factor pricing model is robust to moderate deviations in $\varphi = 0.5$. Therefore, fixing $\varphi = 0.5$ is a well-justified choice that ensures reliable and accurate load forecasting across different pricing scenarios. It should be noted that the optimal value $\varphi = 0.5$ is determined specifically based on the load characteristics and policy environment of Fujian Province. For other provinces or regions with different load patterns or pricing policies, the optimal φ can be re-calibrated following the same methodology presented in this paper.

Table 15
Performance comparison with mainstream forecasting models.

Predictive model	R^2	MAE	RMSE	MAPE (%)
CNN-LSTM	0.918	433.59	542.84	5.45
Transformer	0.935	393.51	476.14	4.86
TCN-GRU	0.949	360.57	421.18	4.34
TCN-LSTM	0.963	316.29	358.36	3.99
Informer	0.983	138.49	167.24	1.79
Proposed model	**0.995**	**74.16**	**99.79**	**0.94**

Table 16
Ablation study results and comparison of denoising strategies.

Predictive model	R^2	MAE	RMSE	MAPE (%)
Without BiGRU	0.933	398.23	483.97	4.92
Without BiTCN	0.954	348.74	401.55	4.18
Without DRM-TD	0.970	274.54	335.52	3.52
Without AMFFS	0.974	264.21	296.26	3.29
Without MHA	0.981	156.42	183.57	1.85
Direct denoising	0.984	132.74	160.71	1.72
Proposed model	**0.995**	**74.16**	**99.79**	**0.94**

4.6. Model comparison and ablation studies

To comprehensively validate the superiority of the proposed BiTCN-BiGRU-MHA architecture and quantify the individual and synergistic contributions of each module, comparative experiments against state-of-the-art forecasting models and systematic ablation studies were conducted. To ensure a fair comparison, all models were implemented under the same data preprocessing pipeline, input feature set, experimental environment, and evaluation metrics. Meanwhile, the hyperparameters of each benchmark model were tuned to their respective optimal configurations based on recommendations from the literature and validation set performance. All models were trained with a standard early stopping mechanism to avoid overfitting. Furthermore, to verify the rationality of implementing multi-modal denoising after VMD decomposition instead of direct denoising on raw data, a dedicated control experiment was established.

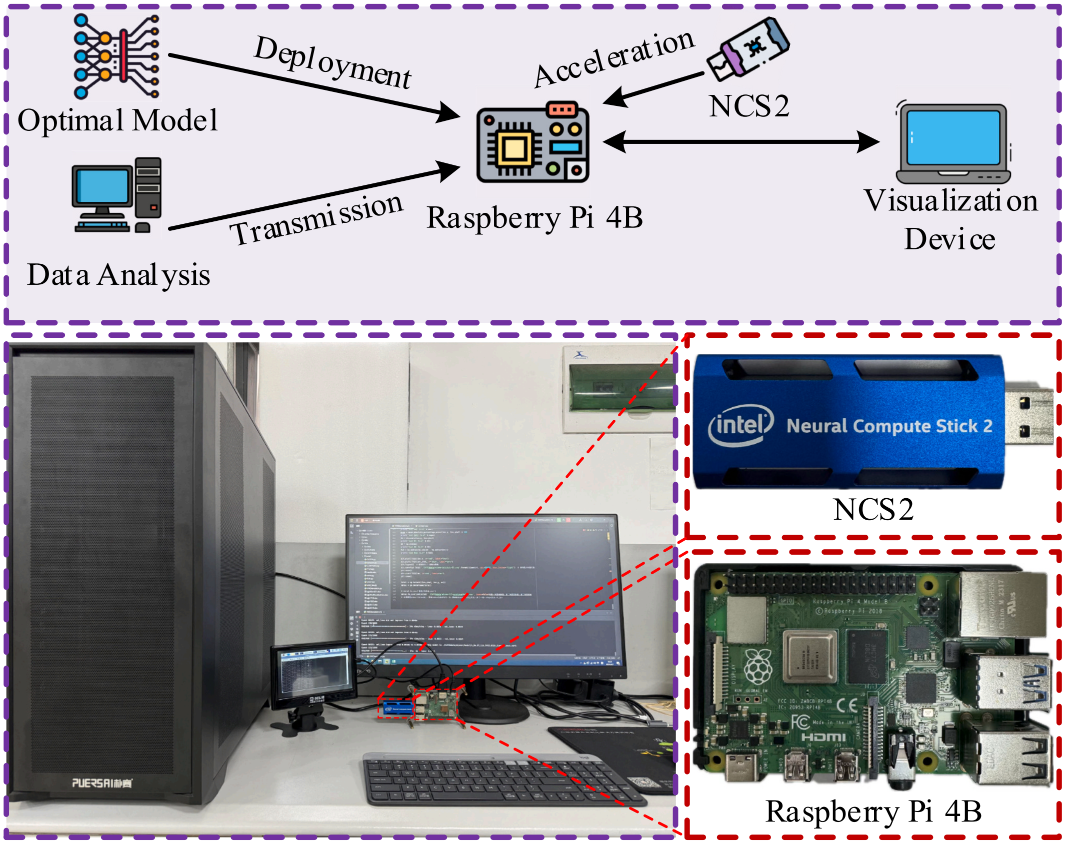


Fig. 19. Simulation verification platform for raspberry Pi.

(1) Comparison with mainstream forecasting models.

The proposed model was compared with several representative deep learning architectures, including CNN-LSTM, Transformer, TCN-GRU, TCN-LSTM, and Informer. All models were evaluated under the identical TOU scenario using the Quanzhou grid dataset. The obtained results are summarized in Table 15.

As shown in Table 15, the proposed model consistently outperforms all benchmark models across all evaluation metrics. Compared to the second-best model, Informer, the proposed model improves R^2 by 1.2%, reduces MAE by 46.4% from 138.49 MW to 74.16 MW, and lowers MAPE by 47.5% from 1.79% to 0.94%. This superior performance can be attributed to the synergistic combination of bidirectional temporal convolution BiTCN for multi-scale feature extraction, bidirectional gated recurrent units BiGRU for capturing long-term dependencies, and the multi-head attention mechanism for dynamic feature weighting.

The limitations of the benchmark models further highlight the advantages of our approach. The CNN-LSTM and Transformer models lack the ability to effectively capture bidirectional temporal dependencies or multi-scale periodicities, resulting in MAPE values above 4.8%. The TCN-GRU and TCN-LSTM hybrids, while using dilated convolutions, still rely on unidirectional recurrent structures, which fail to leverage future context for abrupt load changes induced by price signals. The Informer, though designed for long sequences, simplifies attention distribution for medium-term forecasting and omits bidirectional processing, limiting its sensitivity to price-driven load fluctuations. In contrast, the proposed architecture integrates bidirectional processing in both temporal convolution and recurrent units, coupled with dynamic attention weighting, to achieve superior accuracy and robustness.

(2) Ablation studies and denoising strategy justification.

To quantify the contribution of each key component, systematic ablation experiments were performed by removing one module at a time: the DR model with time delay (DRM-TD), AMFFS denoising, BiGRU, BiTCN, and multi-head attention (MHA). Furthermore, to justify the necessity of performing multi-modal denoising after VMD-based decomposition, an additional variant was introduced in which the AMFFS method was applied directly to raw load data without prior VMD decomposition (denoted as "Direct denoising"). All ablation variants were evaluated under the identical TOU scenario. The corresponding results are presented in Table 16.

The ablation results reveal that removing any module leads to a noticeable degradation in forecasting accuracy, confirming the indispensable role of each component. The most severe performance drop occurs when BiGRU is removed, with R^2 falling to 0.933 and MAPE rising to 4.92%, highlighting the critical importance of bidirectional recurrent processing for capturing temporal dependencies in load data. Removing DRM-TD also causes a substantial increase in MAE from 74.16 MW to 274.54 MW, demonstrating that modeling user response behavior under time-varying pricing is essential for accurate load forecasting. The removal of the multi-head attention mechanism results in a moderate but still significant accuracy loss, indicating that dynamic feature weighting contributes to refining the final predictions.

Table 17
Edge deployment platform parameters.

Equipment type	Hardware Platform	Software platform	Version information
Edge computing device	Raspberry Pi 4B	Python	3.7.0
Model acceleration device	NCS2	OpenVINO	2021.4.752
Edge operating system	Debian buster	VNC viewer	7.6.0

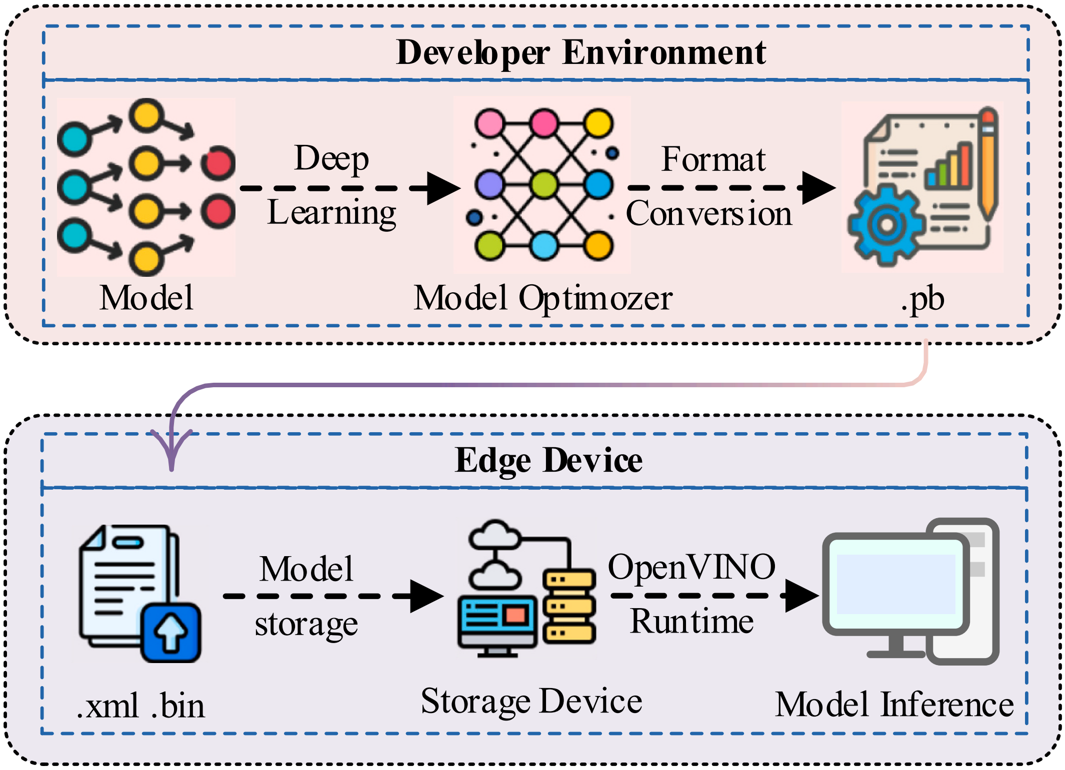


Fig. 20. Systematic model optimization framework leveraging intel OpenVINO toolkit.

Table 18
Model comparison under different electricity pricing mechanisms for edge deployment.

Model implementation	R^2	MAE	RMSE	MAPE (%)	Inference time (s)
TOU					
Unoptimized model (PC simulation)	0.995	74.16	99.79	0.936	4.91
Unoptimized model (raspberry Pi)	0.995	74.05	99.82	0.941	10.12
Optimized model (raspberry Pi)	0.992	106.21	121.82	1.38	3.32
Optimized model (NCS2)	**0.992**	**106.95**	**121.01**	**1.37**	**1.02**
RTP					
Unoptimized model (PC simulation)	0.994	75.21	95.42	1.01	4.89
Unoptimized model (raspberry Pi)	0.994	75.27	95.44	1.01	10.05
Optimized model (raspberry Pi)	0.991	112.45	147.92	1.46	3.28
Optimized model (NCS2)	**0.991**	**112.17**	**147.94**	**1.48**	**1.03**

More importantly, the comparison between direct denoising and the full model provides a clear justification for the proposed strategy of performing multi-modal denoising after VMD-based decomposition. Direct denoising, which applies the AMFFS method directly to the raw load sequence without prior decomposition, achieves an R^2 of 0.984, MAE of 132.74 MW, and MAPE of 1.72%. Although this is superior to the variant without AMFFS, it is markedly inferior to the full model that first decomposes the signal using BOA-VMD and then applies AMFFS to the individual IMF components. The full model reduces MAE by an additional 44.1% from 132.74 MW to 74.16 MW and lowers MAPE by 45.3% from 1.72% to 0.94% compared to direct denoising.

The superiority of decomposition-based denoising stems from the fact that raw power load data simultaneously contain multiple time-scale features: low-frequency trends, such as daily and weekly cycles; medium-frequency periodic components, such as intra-day load patterns; and high-frequency random noise. Applying a single denoising threshold directly on the raw signal inevitably mixes these components, leading to either insufficient noise suppression or over-smoothing of meaningful high-frequency variations. In contrast, BOA-VMD first adaptively separates the signal into distinct IMF with different central frequencies. The subsequent AMFFS method then applies a dynamic, physically constrained threshold to each IMF component individually, preserving the trend and periodic features while effectively eliminating noise. This decompose-then-denoise paradigm enables a more precise decoupling of signal components, which is why the full model significantly outperforms the direct denoising approach. The experimental results thus strongly support the proposed design choice.

In summary, the comparative and ablation experiments collectively validate three key aspects of the proposed method: first, the BiTCN-BiGRU-MHA architecture achieves state-of-the-art forecasting accuracy, outperforming mainstream models by a substantial margin; second, each module including DRM-TD, AMFFS, BiTCN, BiGRU, and MHA makes a distinct and necessary contribution to the overall performance; and third, the decomposition-based multi-modal denoising strategy is essential, as it significantly outperforms direct denoising on raw data.

4.7. Feasibility analysis of model deployment based on raspberry Pi and NCS2

To validate the model's effectiveness in real-world deployment scenarios, this paper deployed an edge computing platform based on the Raspberry Pi 4B. The hardware architecture is illustrated in Fig. 19.

The Raspberry Pi 4B was selected as an ideal edge deployment platform due to its cost-effectiveness, low power consumption, portability, and compatibility with edge computing frameworks. To address the computational demands of real-time temporal data processing in load forecasting applications, the Neural Compute Stick 2 (NCS2) was integrated for inference acceleration. By offloading neural network computations from the host CPU to the dedicated vision processing unit (VPU), the system achieved significant acceleration in inference speed while maintaining a low power envelope, as detailed in Table 17.

To achieve end-to-end acceleration in model optimization and edge deployment, this paper leverages the Intel OpenVINO toolkit, which systematically streamlines the workflow into two interconnected phases. The model optimization phase employs the model optimizer to perform operator fusion and memory optimization, converting the trained model into an intermediate representation (IR) format comprising a topology file (.xml) and associated weight parameters (.bin). Subsequently, during the inference deployment phase, the OpenVINO runtime interface loads the IR files and executes computations on the vision processing unit (VPU) embedded within the NCS2 accelerator, enabling real-time prediction results to be fed back into the power grid dispatch system for decision support. The specific workflow is shown in Fig. 20.

The experiment simulates multiple electricity load curves in various scenarios using the test set, comparing the performance under both TOU and RTP. The evaluation metrics include MAE, RMSE, MAPE, R^2 and average inference speed. A systematic performance comparison is detailed in Table 18.

In the TOU scenario, the unoptimized model shows nearly identical accuracy between the PC simulation and raspberry Pi computations, with R^2 values of 0.995 and minimal differences in MAE, RMSE, and MAPE, indicating that the raspberry Pi, despite being a lightweight edge device, can maintain a similar prediction ability to the PC. However, due to limited hardware capabilities, the inference time on the raspberry Pi is 10.12 s, significantly slower than the 4.91 s on the PC. After optimization with OpenVINO, the inference time on the raspberry Pi is greatly reduced to 3.32 s, showing significant acceleration, though there is a slight loss in prediction accuracy due to operator fusion and memory optimization. When further accelerated with NCS2, the inference time drops to 1.02 s, over 10 times faster than the unoptimized Raspberry Pi computation, with only a minor decrease in accuracy. This demonstrates that NCS2, through its built-in VPU's heterogeneous computing architecture, greatly improves the real-time performance of the edge device while maintaining prediction accuracy.

In the RTP scenario, the unoptimized model on the raspberry Pi takes 10.05 s to perform inference, which is similar to the time in the TOU scenario but still constrained by the edge device's computing power. After optimization with OpenVINO, the inference time on the raspberry Pi reduces to 3.28 s, though the accuracy loss is more pronounced than in the TOU scenario, indicating that the dynamic nature of data features renders the model more sensitive to structural changes. With NCS2 acceleration, the inference time further drops to 1.03 s, with accuracy metrics similar to the optimized raspberry Pi calculations, further confirming the performance boost from hardware acceleration.

The above analysis demonstrates that, whether in TOU or RTP scenarios, NCS2 hardware acceleration significantly reduces inference time while keeping accuracy loss within an acceptable range for engineering applications. The edge deployment solution combining OpenVINO optimization and NCS2 hardware acceleration achieves substantial improvements in inference efficiency without compromising prediction accuracy, providing a viable pathway for deploying load forecasting models on low-power edge devices.

5. Conclusion

To address the challenges posed by nonlinear and highly dynamic load characteristics in VPP scenarios, which existing forecasting methods struggle to handle, this paper proposes an integrated short-term forecasting approach combining BOA-enhanced VMD, AMFFS, and a DR model incorporating user response delay. The proposed data preprocessing approach significantly enhances the performance of deep learning models in load forecasting. By mitigating the interference of random fluctuations, the method enables the model to more accurately capture load variation trends, thereby improving forecasting accuracy. Based on psychological principles, a DR model incorporating time-delay factors is proposed. This model effectively identifies users' dynamic response characteristics under different pricing mechanisms and quantifies response intensity using a logistic function. Although both DRM-TD and AMFFS demonstrate optimization potential across various pricing mechanisms, the DRM-TD model achieves substantially greater error reduction in RTP scenarios, highlighting its superior capability to capture behavior driven by dynamic pricing signals compared to AMFFS.

CRediT authorship contribution statement

Jingzheng Li: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Zhiwen Zhao:** Conceptualization, Data curation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Tao Jin:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Mohamed A. Mohamed:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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